MambaSNN: An Unsupervised Hyperspectral Band Selection Method Based on State-Space Model and Spiking Neural Networks

Zhenyu Wang, Zhongan Tang, Yiliang Wan, Wenzhong Shi, Hua Zhang, Chunhui Li, and Yang Zhao

***Abstract*—Hyperspectral image (HSI) analysis critically relies on band selection (BS) to reduce dimensionality, improve classification accuracy, and minimize computational costs. Despite recent advances in unsupervised deep learning approaches, there exist some issues, including capturing long-range spectral dependencies, which can lead to substantial computational overhead when processing high-dimensional data. To address these challenges, we propose MambaSNN, a novel unsupervised BS method that integrates a state-space spectral module (SpeMamba) with a spiking-neural-network reconstruction module (SNNRecNet). SpeMamba captures intrinsic long-range spectral dependencies via recursive state transitions. This enables efficient and discriminative feature extraction with linear computational complexity. Meanwhile, SNNRecNet embeds leaky integrate-and-fire (LIF) spiking neurons into a convolutional reconstruction network, encouraging sparse spike-based encodings and alleviating the impact of noisy information, thereby preserving more discriminative spectral spatial information for BS. By jointly optimizing these modules, MambaSNN can effectively reduce spectral redundancy while maintaining classification performance. Comprehensive experiments have been conducted to validate the effectiveness of the proposed MambaSNN across five benchmark HSI datasets. Comparative results demonstrate that MambaSNN consistently outperforms state-of-the-art BS methods in both accuracy and runtime efficiency. These results validate the efficacy and practicality of the proposed approach for unsupervised HSI classification in high-dimensional, resource-constrained applications.**

***Index Terms*—Hyperspectral image (HSI), Mamba, spiking neural network (SNN), state-space model (SSM), unsupervised band selection (BS).**

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Zhenyu Wang, Yiliang Wan, and Yang Zhao are with the School of Geographic Sciences, Hunan Normal University, Changsha 410081, China, and also with Hunan Provincial Key Laboratory of Geospatial Big Data Mining and Application, Changsha 410081, China (e-mail: wang0921@hunnu.edu.cn; wanylir@hunnu.edu.cn; zhaoyang@hunnu.edu.cn).

Zhongan Tang and Chunhui Li are with The Third Surveying and Mapping Institute of Hunan Province and Hunan Engineering Research Center of Geographic Information Security and Application, Changsha 410000, China (e-mail: whtza@163.com; 18570305782@163.com).

Wenzhong Shi is with the Department of Land Surveying and Geo-Informatics, Smart Cities Research Institute, The Hong Kong Polytechnic University, Hong Kong, China (e-mail: lswzshi@polyu.edu.hk).

Hua Zhang is with the School of Environment and Spatial Informatics, China University of Mining and Technology, Xuzhou 221116, China (e-mail: zhhua_79@163.com).

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I. Introduction

HYPERSPECTRAL image (HSI) has garnered substantial attention in remote sensing and image processing research owing to its significance in diverse applications, including land cover classification [1], [2], mineralogical exploration [3], and medical imaging [4], [5]. Hundreds of contiguous spectral bands are characteristic of HSI, which delivers rich spectral and spatial information. This, in turn, confers considerable advantages for image classification and analytical tasks [6], [7]. Nonetheless, the high-dimensional nature of such data introduces significant challenges, including the Hughes phenomenon, substantial information redundancy, and prohibitive computational demands [8], [9]. Therefore, dimensionality reduction has been an essential preprocessing imperative.

Dimensionality reduction in HSI is typically achieved through either feature extraction or band selection (BS). Feature extraction methods, such as principal component analysis (PCA) [10] and linear discriminant analysis (LDA) [11], reduce dimensionality by projecting data into lower-dimensional subspaces. Although these methods are effective, they alter the original spectral structure and often compromise the physical interpretability of the data [12]. In contrast, BS methods aim to directly identify the most informative spectral bands. These approaches preserve the inherent physical properties of HSI and are more conducive to practical deployment [13], [14]. Generally, BS methods can be categorized into supervised, semi-supervised, and unsupervised paradigms [15]. Supervised and semi-supervised methods rely on labeled samples for model training [16], [17]. However, acquiring labeled hyperspectral data is often labor-intensive and expensive, which limits the applicability of these methods. Consequently, unsupervised BS methods, which operate without labeled data, have garnered increasing interest. These methods exploit statistical or structural properties of HSI to

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select informative bands, thereby reducing redundancy and improving computational efficiency [18], [19], [20].

Conventional unsupervised BS methods are commonly grouped into three categories: ranking-based [21], [22], search-based [23], [24], and clustering-based methods [25]. Ranking-based methods compute metrics such as variance, entropy, or mutual information for individual bands and prioritize the top-ranked bands for selection. Examples of this category include maximum variance PCA (MVPCA) [26] and manifold ranking-based selection [27]. These methods are computationally efficient and noniterative but often fail to consider interband correlations, which may result in redundant selection. Search-based techniques formulate BS as a combinatorial optimization problem and aim to identify optimal band subsets using strategies such as adaptive particle swarm optimization (APSO-BS) [28], multiobjective cuckoo search (MOCS) [29], and robust dual graph self-representation (RDGSR) [30]. Although these approaches balance information preservation and redundancy reduction, they typically entail high computational costs and are prone to local optima. Clustering-based methods partition spectral bands into groups and select representative bands from each cluster. Techniques such as k-means, spectral clustering, and density peak clustering are commonly used. Notable examples include the difference between intergroups (DIG) method [31] and hierarchical latent feature representation learning-guided clustering (HLFC) [32]. While clustering approaches promote spectral diversity, they often require prior knowledge of cluster numbers and tend to be unstable under noisy or complex spectral conditions.

To address the limitations of traditional BS methods, deep learning-based BS approaches have received increasing attention. Convolutional neural networks (CNNs) are frequently adopted to capture spatial–spectral characteristics and improve BS performance [33], [34]. However, due to the limited receptive fields of CNNs, their ability to model long-range dependencies among distant spectral bands is inherently constrained. In comparison, graph neural networks (GNNs) exhibit superior capability in modeling global dependencies, as they construct relational graphs to capture complex interband interactions [35], [36]. Nevertheless, GNN-based methods rely on iterative neighborhood aggregation, a process that introduces significant computational overhead and may hinder real-time applications. Transformer-based models further enhance the modeling of global relationships by employing self-attention mechanisms [37], [38], [39]. Despite their effectiveness, these models suffer from quadratic complexity, which scales poorly with the high dimensionality of HSI and leads to computational intractability in practice.

Furthermore, to enable more effective spectral feature extraction via deep learning, an increasingly adopted paradigm in BS research involves the integration of reconstruction networks as an optimization backbone. These frameworks leverage reconstruction objectives, enabling the refinement of spectral features through end-to-end training via backpropagation (BP). Consequently, the representational quality of the selected bands is enhanced. This paradigm has been implemented in a number of notable ways. One such example is convolution–deconvolution architectures (Conv–Deconv Nets), such as the 2D-CNN-based BS-Net-Conv [40], the 1D-CNN-based global–local spectral weight network with attention (GLSWA) [41], and 3D-CNN-based models like the dual-attention reconstruction network (DARecNet) [42] and spectral–spatial cross-dimensional attention network (SSANet-BS) [43]. However, models that are exclusively reliant on convolutional operations have been shown to exhibit protracted convergence, unstable outcomes, and suboptimal reconstruction fidelity, thereby constraining their efficacy in practical applications.

Motivated by these challenges, this article proposes MambaSNN, a novel unsupervised BS reconstruction network that adopts the Mamba architecture as the spectral backbone. Mamba is a selective state-space model (SSM) that processes sequences through recurrent state transitions while maintaining linear complexity with respect to sequence length. Compared with transformer-based methods, Mamba can model long-range spectral dependencies over hundreds of bands with significantly reduced computational and memory overhead. At the same time, it offers a stronger capacity for global dependency modeling than conventional CNNs, whose receptive fields grow only gradually with depth. Therefore, we propose a Mamba-based BS method named MambaSNN for addressing the tradeoff between long-range spectral modeling and computational costs. The proposed MambaSNN consists of two main components: a spectral feature extraction module (SpeMamba) based on Mamba SSM blocks, and a spiking-neural-network reconstruction module (SNNRecNet) built upon spiking neural networks (SNNs). SpeMamba encodes hyperspectral sequences via recursive state transitions, efficiently capturing global interband relationships and extracting discriminative spectral features under a linear-time complexity regime. SNNRecNet then exploits leaky integrate-and-fire (LIF) neuronal dynamics within a convolutional framework to reconstruct the full spectral cube from the selected bands. The inherent sparsity of SNN activations promotes efficient and stable reconstruction, further reducing redundancy in the learned features. MambaSNN attains consistent accuracy gains over algebraic BS methods at the cost of moderate additional runtime, while remaining more efficient than existing deep learning-based BS models. Mamba is well-suited for hyperspectral BS by efficiently modeling long-range spectral dependencies, while SNNs are well-suited for reconstruction due to their event-driven sparse activations and biologically inspired neuronal dynamics, which can suppress redundant responses and improve robustness in feature recovery. This combination can enable MambaSNN to achieve long-range spectral modeling, linear computational complexity $\mathcal{O}(N)$, and robust reconstruction. These achievements contribute to neuromorphic hyperspectral processing, while simultaneously advancing unsupervised hyperspectral feature learning and BS.

In summary, the principal contributions of this work are as follows.

1) We propose MambaSNN, an unsupervised BS method for HSI-based land cover classification that jointly optimizes BS quality and computational efficiency.

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2) The SpeMamba module serves as a Mamba-based spectral encoder that models long-range interband dependencies with linear complexity $\mathcal{O}(N)$, extracting globally discriminative hyperspectral features.
3) The SNNRecNet module serves as a spiking reconstruction network that exploits sparse neuronal dynamics to recover spectral information from selected bands, enhancing BS accuracy and model stability.
4) Experimental evaluations across diverse scenarios demonstrate that the MambaSNN method surpasses state-of-the-art methods in computational efficiency and BS accuracy.

The remainder of this manuscript is organized as follows. Section II delineates foundational concepts of SSMs and SNNs. Section III details the proposed MambaSNN methodology for unsupervised hyperspectral BS. Section IV describes the experimental framework employed to evaluate the proposed method, compares its performance against state-of-the-art BS techniques, and presents a comprehensive results analysis. Concluding remarks are provided in Section V.

II. Preliminary

A. State-Space Model

Inspired by linear time-invariant system principles, SSM [44] transforms a 1-D input function or sequence $x(t) \in \mathbb{R}$ into an output $y(t) \in \mathbb{R}$ via an intermediate latent state $h(t) \in \mathbb{R}^N$. This transformation is governed by a linear ordinary differential equation (ODE)

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ denotes the state transition matrix, while $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{1\times N}$ represent input and output projection matrices, respectively.

To adapt this continuous formulation for discrete sequence processing within deep learning architectures, SSM incorporates a time-scale parameter Δ and employs zero-order hold (ZOH) discretization [45] of continuous parameters $\mathbf{A}$ and $\mathbf{B}$

$$\begin{aligned} \bar{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \bar{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right) \cdot \Delta\mathbf{B} \end{aligned} \tag{2}$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ signify the discretized counterparts of $\mathbf{A}$ and $\mathbf{B}$.

The discretized SSM system is consequently expressed as

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{3}$$

Computational efficiency is achieved by implementing this linear recurrence through convolutional operations

$$\begin{aligned} \bar{\mathbf{K}} &= \left(\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}\right) \\ \mathbf{y} &= \mathbf{x} * \bar{\mathbf{K}} \end{aligned} \tag{4}$$

where L denotes the input sequence length and $\bar{\mathbf{K}} \in \mathbb{R}^L$ represents the structured convolution kernel.

Conventional SSM exhibits limitations in contextual relationship modeling due to static parameterization. To address this constraint, the Mamba architecture [46] introduces a selection mechanism via the S6 module, enabling dynamic state interactions through sequence-aware parameterization

$$\begin{aligned} \mathbf{B} &= \mathcal{P}_B(\mathbf{x}) \\ \mathbf{C} &= \mathcal{P}_C(\mathbf{x}) \\ \Delta &= \tau_\Delta\left(\Theta + \mathcal{P}_\Delta(\mathbf{x})\right). \end{aligned} \tag{5}$$

Here, parameters $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are dynamically generated from input $\mathbf{x} \in \mathbb{R}^{B\times L\times D}$, facilitating contextually adaptive selection for individual sequence elements.

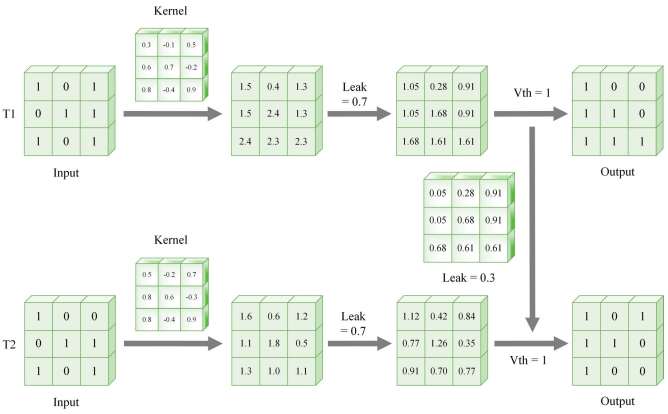


Fig. 1. Computational schematic of simplified spike convolution across two temporal iterations, illustrating 2-D input and kernel operations.

B. Spiking Neural Networks

Based on the computational principles underlying biological neural systems, recent advances in neuromorphic computing have sought to emulate cortical processing through SNNs [47], [48]. Spike train processing is accomplished via spiking convolutional layers, with the operational mechanics illustrated in Fig. 1.

The present framework employs LIF neuronal dynamics [49], [50]. At each convolutional epoch, spike sequences undergo membrane potential leakage governed by a decay constant. At temporal iteration $t = 1$, the previous neuronal spike train undergoes convolution with the synaptic kernel, and the resulting postsynaptic activation is accumulated in the subsequent neurons. When this activation surpasses the membrane potential threshold V_{th}, an output spike sequence is generated. Otherwise, subthreshold activations persist and summate with subsequent temporal inputs. This biophysically inspired process is formalized as

$$\begin{aligned} v_i^t &= \left(1 - \tau_m'\right) v_i^{(t-1)} + \tau_m' \sum w_{ij} o_j^t, \quad \tau_m' \in (0 \sim 1) \\ o_j^t &= \begin{cases} 0, & v_j^t < V_{th} \\ 1, & v_j^t \geq V_{th} \end{cases} \end{aligned} \tag{6}$$

where τ_m' denotes the membrane potential decay coefficient, controlling the rate of potential leakage; w_{ij} represents the synaptic efficacy between neurons i and j; v_i^t signifies the membrane potential of neuron i at iteration t; and $o_i^t \in \{0, 1\}$ indicates whether a spike is emitted. The forward propagation

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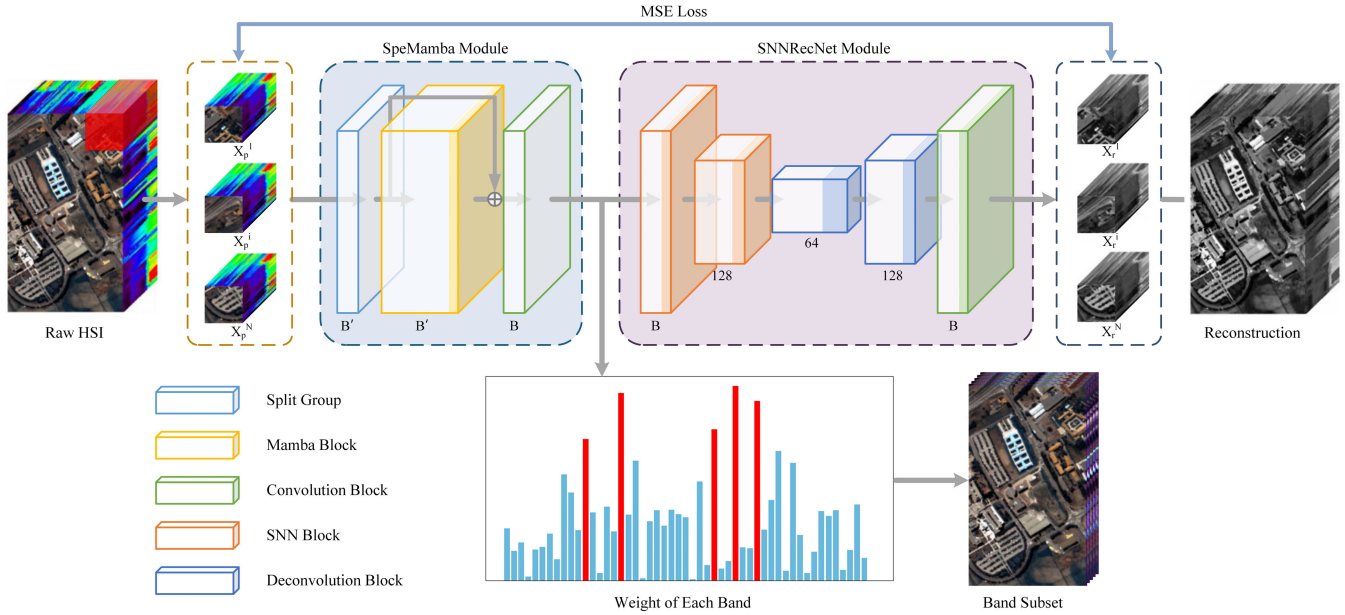


Fig. 2. Overview of the proposed MambaSNN framework.

potential v_i^t integrates the residual potential v_i^{t-1} from prior iteration $t-1$ with current presynaptic activations o_j^t, modulated by the decay mechanism.

In (6), suprathreshold membrane potentials ($v_j^t \geq V_{th}$) trigger binary spike emissions ($o_j^t = 1$), while subthreshold potentials maintain quiescence ($o_j^t = 0$). Following spike emission, the neuron's membrane potential is instantaneously reset to its resting value, establishing refractory dynamics until new inputs are integrated.

III. Methodology

A. MambaSNN: Overview

The comprehensive architecture of the proposed MambaSNN method is illustrated in Fig. 2. This integrated system comprises two principal components: the SpeMamba module and the SNNRecNet module. The SpeMamba module, together with the S6 mechanism, is designed to simultaneously capture intragroup (local) correlations that reflect spectral regionalization and intergroup (long-range) dependencies across distant bands in a unified and efficient framework. The SNNRecNet module implements a convolutional encoder-deconvolutional decoder structure, executing a compression-restoration pipeline to achieve image reconstruction. Global optimization is accomplished by minimizing the mean-square error (MSE) between the reconstructed output and the original imagery. Ultimately, BS emerges from the synergistic integration of extracted spectral features and the reconstruction-optimized representations. This process yields band-specific weighting coefficients that identify the most informative band features.

B. SpeMamba Module

The SpeMamba in the proposed MambaSNN acts as a spectral feature extractor, designed to extract and enhance discriminative spectral signatures from raw hyperspectral imagery. Consider the input hyperspectral data tensor $\mathbf{X} \in \mathbb{R}^{B\times W\times H}$, where B denotes the spectral band count, while W and H represent spatial dimensions. To ensure numerical stability and unify spectral magnitudes, $\mathbf{X}$ is first subjected to min–max normalization along the spectral dimension. This process ensures that the spectrum of each pixel is scaled to the range [0, 1]. To capture local spectral–spatial correlations, SpeMamba employs a sliding–window sampling strategy. This involves an $m \times m$ rectangular window that moves with stride t along the spatial axes and systematically extracts N overlapping patches $\mathbf{X}_p \in \mathbb{R}^{B\times m\times m}$. Each patch thus preserves the full spectral resolution while localizing spatial context. In order to facilitate subsequent spectral grouping, channel dimension padding transforms $\mathbf{X}_p$ into $\mathbf{X}_g \in \mathbb{R}^{B'\times m\times m}$, where $B' = \lceil B/G \rceil \times G$ and G signifies the spectral group cardinality.

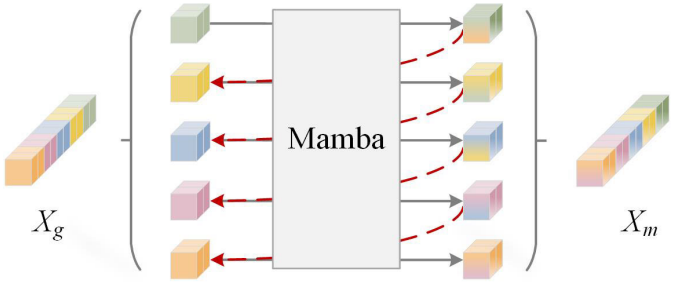


Fig. 3. Mamba enhances the extraction of spectral features by fusing spectral sequence information.

During feature transformation, each spectral patch is partitioned into G feature groups (see Fig. 3). This partitioning strategy not only reduces computational complexity but also, and more importantly, lays the groundwork for capturing intergroup dependencies. By structuring the spectral information in this way, the model can effectively learn how different spectral groups interact with one another. Through the integration of the S6 mechanism, the MambaSNN

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dynamically adapts its attention to the relationships between these groups, modulating the focus of feature extraction according to the importance of intergroup interactions. This dynamic and context-aware modulation enables more precise and adaptive feature extraction, enhancing the network's ability to capture intricate spectral relationships that are critical for accurate hyperspectral analysis.

The transformation pipeline in SpeMamba comprises three sequential stages that progressively enhance the spectral features. First, spectral patches undergo feature extraction via the Mamba module, which captures salient spectral–spatial patterns within the HSI. Subsequently, the extracted features are passed through group normalization (GN) followed by a sigmoid linear unit (SiLU) activation function, which can provide the nonlinear expressiveness needed for high-performance discriminative models. Finally, residual connections are employed to combine the processed features with the original spectral inputs through a convolutional fusion layer. This results in an enriched, aggregated feature representation. The transformation process is formally expressed as

$$\begin{aligned}\mathbf{Z} &= \text{Mamba}\left(\mathbf{X}_g\right)\\ \mathbf{X}_m &= \text{SiLU}\left(\text{GN}\left(\mathbf{Z}\right)\right)\\ \mathbf{X}'_p &= \text{Conv}\left(\mathbf{X}_m \oplus \mathbf{X}_g\right)\end{aligned} \tag{7}$$

where $\mathbf{Z}, \mathbf{X}_m \in \mathbb{R}^{B' \times m \times m}$ share the same shape as the input $\mathbf{X}_g$. The operator Mamba$(\cdot)$ implements the selective SSM described in Section II-A, and $\text{SiLU}(x) = x \cdot \sigma(x)$ denotes the SiLU activation function. The output $\mathbf{X}'_p \in \mathbb{R}^{B \times m \times m}$ is obtained by a 1×1 convolutional fusion layer, which projects the B' grouped channels back to B spectral channels. The residual connection $\oplus$ adds the original input $\mathbf{X}_g$ to the normalized and activated features $\mathbf{X}_m$, mitigating gradient attenuation during BP and preserving the integrity of the original spectral information. Here, $\oplus$ denotes element-wise residual addition, ensuring direct feature enhancement without dimensional expansion.

C. SNNRecNet Module

The SNNRecNet module implements a spiking-aware encoder–decoder structure tailored for HSI reconstruction. As shown in Fig. 2, this module integrates convolutional encoders, spiking neuronal activation, and deconvolutional decoders to effectively compress hyperspectral data and then progressively restore the spatial–spectral structure of the input.

The encoder comprises two cascaded spiking convolutional stages, which progressively compress the enhanced HSI block $\mathbf{X}'_p$ into a compact spatiotemporal spike-based representation. Initially, features derived from preceding convolutional layers are systematically transformed through dual 3×3 spiking convolutional operations. Subsequently, LIF neuronal activation dynamics are employed to emulate biological sparsity mechanisms, thereby enhancing nonlinear representational capacity. The encoder is expressed as follows:

$$\begin{aligned}\mathbf{X}s_1 &= \text{LIF1}\left(\text{SConv1}\left(\mathbf{X}'_p\right)\right)\\ \mathbf{X}s_2 &= \text{LIF2}\left(\text{SConv2}(\mathbf{X}s_1)\right)\end{aligned} \tag{8}$$

where $\mathbf{X}s_1 \in \mathbb{R}^{B \times m \times m}$ and $\mathbf{X}s_2 \in \mathbb{R}^{128 \times m \times m}$ denote intermediate compressed representations.

The decoder adopts a symmetric topology, wherein transposed convolutions progressively restore spatial resolution and spectral fidelity. Specifically, the intermediate spike tensor $\mathbf{X}s_2$ is upsampled through two successive stages

$$\begin{aligned}\mathbf{X}t_1 &= \text{ReLU}\left(\text{BN}\left(\text{TransConv1}(\mathbf{X}s_2)\right)\right)\\ \mathbf{X}t_2 &= \text{ReLU}\left(\text{BN}\left(\text{TransConv2}(\mathbf{X}t_1)\right)\right)\end{aligned} \tag{9}$$

where $\mathbf{X}t_1 \in \mathbb{R}^{64 \times m \times m}$ and $\mathbf{X}t_2 \in \mathbb{R}^{128 \times m \times m}$ represent upsampled feature tensors.

The decoding process culminates with a terminal 1×1 convolutional layer that restores the original spectral dimensionality. This operation is subsequently followed by a Sigmoid activation function. It constrains the output within the normalized range of [0, 1], ensuring consistency with the input data scale. Formally, the reconstruction is expressed as

$$\mathbf{X}_r = \text{Sigmoid}\left(\text{Conv}\left(\mathbf{X}t_2\right)\right) \tag{10}$$

where $\mathbf{X}_r \in \mathbb{R}^{B \times m \times m}$ constitutes the reconstructed hyperspectral patch, faithfully approximating the original input in both spectral and spatial domains.

D. Training and Optimization

The MambaSNN is trained end-to-end by minimizing the mean squared reconstruction error between the original hyperspectral input patches $\mathbf{X}_p$ and their decoder outputs $\mathbf{X}_r$. Let N denote the number of training patches. The loss function $\mathcal{L}$ is formally defined as

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{X}_p^i - \mathbf{X}_r^i\right\|^2. \tag{11}$$

The stochastic gradient descent (SGD) method is employed for the parameter optimization in the MambaSNN. In each training iteration, forward propagation produces the reconstruction output. Concurrently, the backward propagation updates network parameters θ. However, the discrete spike train representations generated by LIF neurons can prevent direct application of conventional BP algorithms. To overcome this limitation, critical SNN parameters are optimized using the surrogate gradient method [49], which approximates the nondifferentiable LIF dynamics with smooth surrogate derivatives.

Following parameter optimization, spectral significance is quantified by computing, for each Mamba-derived feature tensor $\mathbf{X}_p$, a spatial mean activation vector $\mathbf{w}_p \in \mathbb{R}^{B \times 1}$

$$\mathbf{w}_p = \frac{1}{m^2}\sum_{i=1}^{m}\sum_{j=1}^{m}\mathbf{X}_p[i,j]. \tag{12}$$

ReLU activation is then applied to yield band-wise importance weights $\bar{\mathbf{w}}_p \in \mathbb{R}^{B \times 1}$ for each patch during training. To determine global band significance, these per-patch weights are aggregated across all N patches, and the top-ranked b bands are selected according to the following equation:

$$\mathcal{B}_s = \arg\max\left(\sum_{p=1}^{N}\bar{\mathbf{w}}_p, b\right) \tag{13}$$

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Algorithm 1 MambaSNN

Input: Hyperspectral cube $\mathbf{X}$, band subset size b, and MambaSNN hyperparameters.
Output: Final band subset $\mathcal{B}_s$.
Normalize $\mathbf{X}$ and generate N overlapping patches $\mathbf{X}_p$.
for $p = 1, 2, ..., N$ **do**
  Pad and group $\mathbf{X}_p$ to obtain $\mathbf{X}_g$.
  Extract spectral features through Mamba by Eq. (7).
  Reconstruct the HSI through SNN by Eqs. (8) and (9).
  Get the final reconstructed HSI patch by Eq.-DEL (10).
  Calculate reconstruction loss by Eq. (11).
  Backpropagate $\mathcal{L}$ and update network parameters.
  Calculate current channel weight vector $\mathbf{w}_p$ by Eq. (12).
end for
Aggregate channel weights over all patches.
Select top-b bands to form the final band subset $\mathcal{B}_s$ by Eq. (13).
Return The final band subset $\mathcal{B}_s$.

where $\mathcal{B}_s$ denotes the final band subset and b specifies the cardinality of selected bands.

Algorithm 1 formalizes the MambaSNN implementation.

IV. Experiment

A. Datasets

We evaluated our method on five benchmark HSI datasets: the Indian Pines (IP) [51], Pavia University (PU) [52], Houston (H) [53], WHU-Hi-HanChuan (HC) [54], and Toulouse (TLS) [55], [56] datasets. The overview of the five datasets is shown in Table I.

1) *IP:* This scene was acquired over the IP test site in northwest Indiana, containing 145×145 pixels and 220 spectral bands with wavelengths ranging from 0.4 to 2.5 μm. The scene includes 16 land cover classes. Due to the removal of 20 water absorption bands, namely, [104, 108], [150, 163], and [220, 200] spectral bands were retained for the experiment.
2) *PU:* The PU dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) over the University of Pavia, Italy, with a wavelength range of 430–860 nm. The spatial resolution of the scene is approximately 1.3 m, and its size is 610×340, containing nine ground-truth classes. After the removal of twelve noisy bands, 103 spectral bands were retained.
3) *H:* The H dataset was collected by the compact airborne spectral imaging system ITRES CASI-1500 in 2012, covering the campus of the University of Houston. It comprises 349×1905 pixels and 144 spectral bands between 380 and 1050 nm. There are 15 labeled land cover classes in this scene.
4) *HC:* The HC dataset was captured in Hanchuan, Hubei Province, China, using a Headwall Nano Hyperspec sensor mounted on a Leica Aibot X6 UAV. The scene covers a rural–urban fringe area, including buildings, water, and croplands with seven crop types. It consists of 1217×303 pixels and 274 spectral bands ranging from 400 to 1000 nm, with a spatial resolution of approximately 0.109 m.
5) *TLS:* The TLS dataset was acquired by the AisaFENIX sensor during the CAMCATT AI4GEO campaign over Toulouse, France. It contains 420 spectral bands covering 0.4–2.5 μm. A subset of the original image data with a spatial resolution of 1 m was used in the experiment, which includes 14 land cover classes.

B. Experimental Setup

1) Comparative Methodologies: To rigorously assess the performance of MambaSNN, we conducted comparative experiments against seven state-of-the-art unsupervised BS algorithms: MVPCA [26], HLFC [32], RDGSR [30], DARecNet [42], DropoutCAE [57], CapUBS [58], and DISGT [39]. In addition, a full-spectrum baseline was included to quantify the benefit of BS. Table II summarizes the hyperparameter configurations employed for each method across the IP, PU, H, HC, and TLS datasets.

All experiments were performed on a 64-bit Windows 11 platform equipped with an Intel Core i7-10750H CPU and an NVIDIA Quadro RTX 3000 GPU with 6 GB of dedicated memory. The proposed MambaSNN method was implemented in Python and developed using the PyTorch library. Specifically, for the IP, PU, and H datasets, each experiment was trained for 30 epochs with a fixed learning rate of 0.001.

2) Evaluation Metrics: The primary criterion for assessing BS efficacy is classification accuracy on the reduced spectral subset, in accordance with standard evaluation protocols. Two distinct classifiers were employed: a support vector machine (SVM) with a radial basis function kernel [59] and a k-nearest neighbors (KNNs) classifier ($k = 3$) [60]. To ensure statistical robustness, 10% of each class's samples were randomly partitioned for training, with the remainder constituting the test set. Performance quantification employs three standard metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). Each method was executed over ten independent trials with fresh random splits, and we reported the mean of each metric to mitigate the effects of sampling variability.

C. Results Analysis

Figs. 4–8 show the OA, AA, and Kappa obtained from different BS methods and by the full-spectrum baseline under both KNN and SVM classifiers for the five different datasets, respectively. The band subset size is set from 5 to 35 with an increment of 5.

On PU, the proposed MambaSNN consistently outperforms competing methods under both KNN and SVM (see Fig. 4), across varying band subset cardinalities under both KNN and SVM classifiers. Beyond 20 bands, most algorithms (with the exception of MVPCA) converge, exhibiting only marginal differences. However, under constrained spectral budgets, particularly when fewer than 15 bands are selected, MambaSNN shows notable advantages. At 10 bands with KNN, it attains an OA of 87.52% and AA of 85.13%. These correspond to improvements over DISGT, HLFC, and CapUBS

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TABLE I
SUMMARY OF HSI DATASET

Dataset	Sensor	Pixels	Bands	Classes	Spectral Range (μm)	Spatial Resolution (m)
IP	AVIRIS	145×145	200	16	0.4-2.5	20
PU	ROSIS	610×340	103	9	0.43-0.86	1.3
H	CASI	349×1905	144	15	0.38-1.05	2.5
HC	Headwall Nano	1217×303	274	16	0.4-1.0	0.109
TLS	AisaFENIX	1376×1480	420	14	0.4-2.5	1.0

TABLE II
HYPERPARAMETER SETTINGS FOR DIFFERENT BS METHODS

Methods	IP	PU	H	HC	TLS
MVPCA	-				
HLFC	d = 5, T = 750				
DARecNet	η = le-2, maxiter = 30				
DropoutCAE	η = le-3, τ_0 = 1, τ_c = 0.001, C = 40, B = 1				
RDGSR	S = 100, λ = 0.01	S = 500, λ = 10	S = 1000, λ = 0.01	S = 15, λ = 1	S = 500, λ = 0.01
CapUBS	P = 8, α = 1000	P = 10, α = 10	P = 6, α = 100	P = 8, α = 1	P = 10, α = 1000
DISGT	S = 3, σ = 0.5, α = 0.1	S = 2, σ = 0.5, α = 0.1	S = 3, σ = 0.5, α = 0.1	S = 7, σ = 0.5, α = 0.1	S = 3, σ = 0.5, α = 0.1
MambaSNN	G = 4, τ'_m = 0.7	G = 2, τ'_m = 0.7	G = 8, τ'_m = 0.7	G = 16, τ'_m = 0.7	G = 16, τ'_m = 0.7

Note: For MVPCA, the principal component dimension L is set to the number of spectral bands for each dataset.

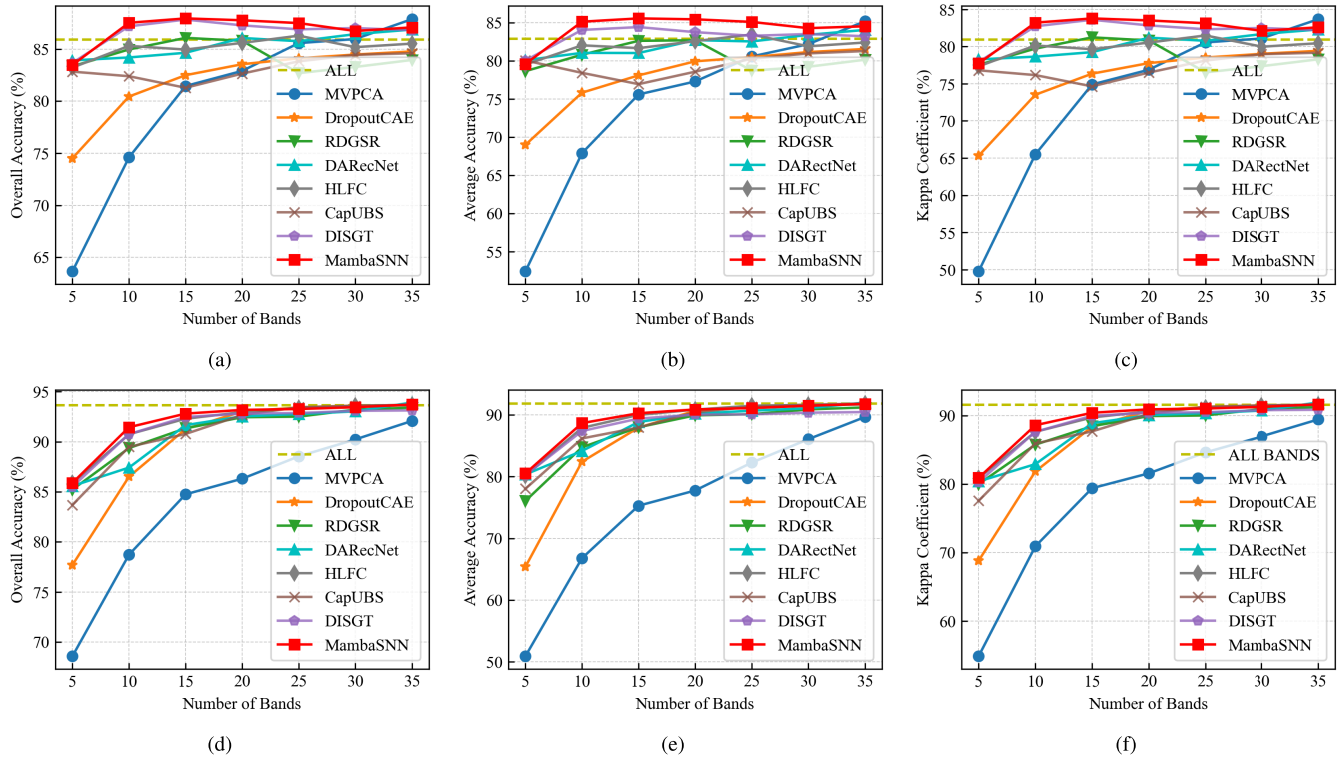


Fig. 4. Classification performance of different BS methods under different band subset sizes on PU dataset. (a)–(c) KNN classification OA, AA, and Kappa results based on different BS methods for the selected bands. (d)–(f) SVM classification OA, AA, and Kappa results.

of 0.34%, 2.22%, and 5.13% in OA, and 1.08%, 3.10%, and 6.71% in AA, respectively. This discriminative capability is further supported by the Kappa, where MambaSNN records 83.21%, outperforming DISGT by 0.49%, HLFC by 3.07%, and CapUBS by 7.05%. Under SVM classification with the same band count, MambaSNN maintains superiority, achieving 91.45% OA and 88.66% AA, with improvements of 0.67% and 1.28% over DISGT, respectively. These findings collectively substantiate the enhanced efficacy of MambaSNN for BS, especially in scenarios with limited spectral resources.

As shown in Fig. 5, MambaSNN demonstrates strong discriminative power on the IP dataset. A notable exception is the extremely low-band regime (5–10 bands), where its accuracy is slightly lower than some other methods, such as HLFC. This deviation is mainly attributable to the IP dataset's homogeneous agricultural classes with subtle interclass spectral differences. In such vegetation-dominated scenes, aggressive band sparsification can increase confusion among similar crop categories, indicating a limitation under semantically uniform land cover settings. As more bands are selected, this kind

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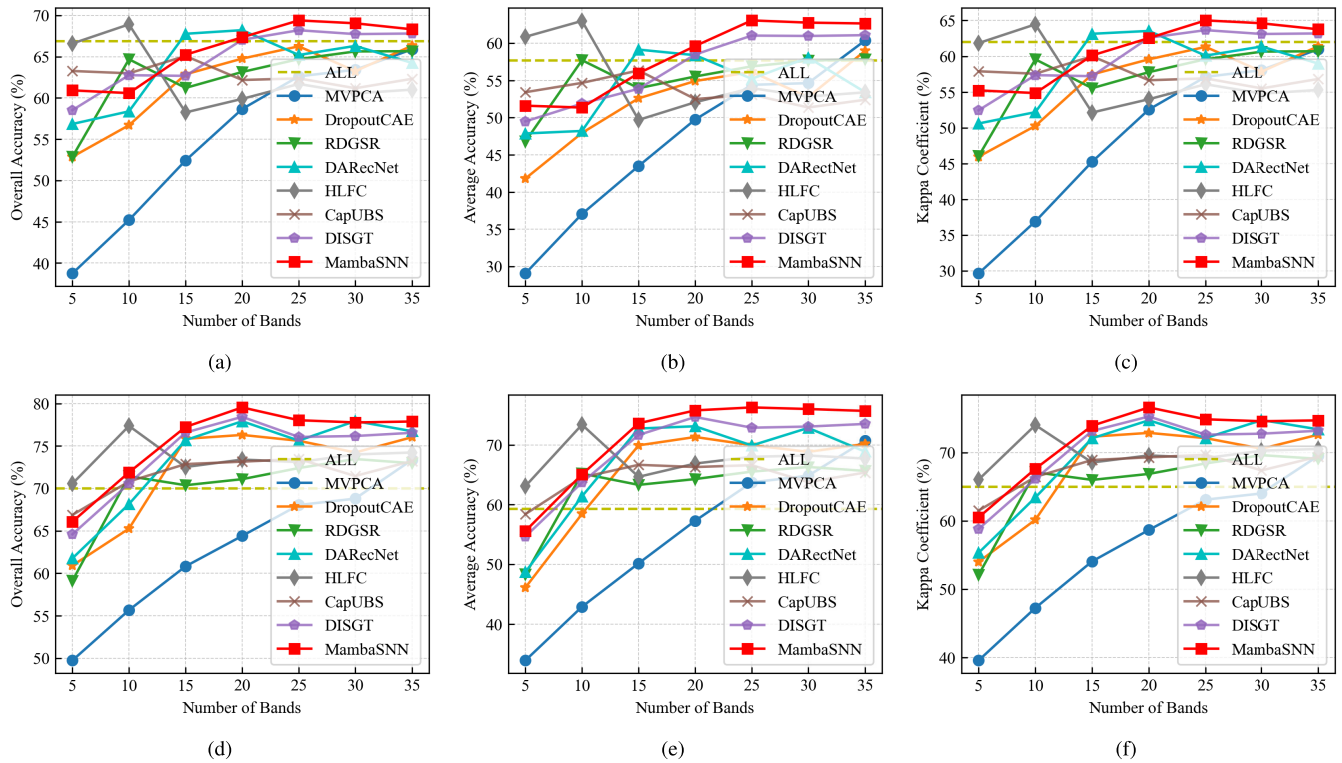


Fig. 5. Classification performance of different BS methods under different band subset sizes on IP dataset. (a)–(c) KNN classification OA, AA, and Kappa results based on different BS methods for the selected bands. (d)–(f) SVM classification OA, AA, and Kappa results.

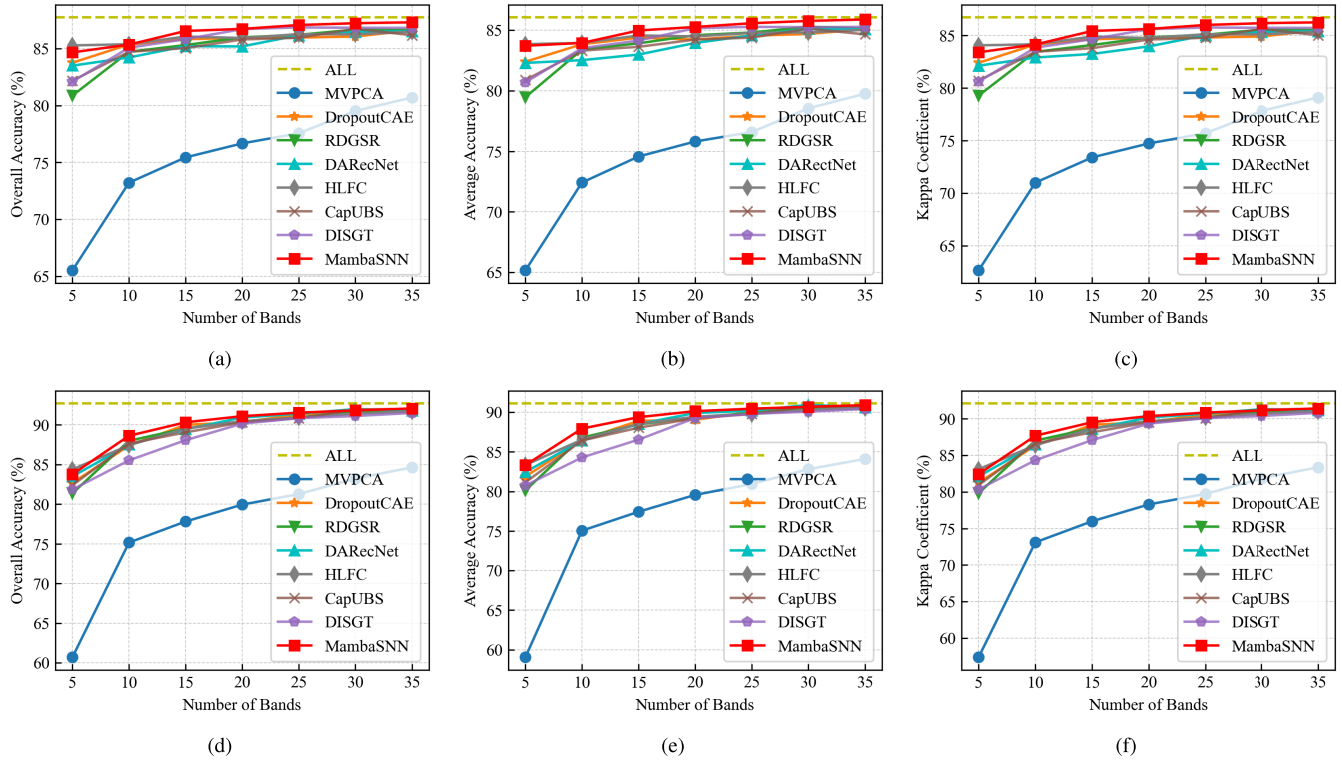


Fig. 6. Classification performance of different BS methods under different band subset sizes on H dataset. (a)–(c) KNN classification OA, AA, and Kappa results based on different BS methods for the selected bands. (d)–(f) SVM classification OA, AA, and Kappa results.

of limitation can be gradually alleviated. At the 20-band setting under SVM, MambaSNN achieves 79.54% OA and 75.80% AA, outperforming the second-best DISGT by 1.11% OA and surpassing the full-band result by 9.53% in OA and 16.48% in AA. Across BSs ranging from 5 to 35, the variability in OA achieved by MambaSNN is minimal. This indicates that the proposed method exhibits superior stability in classification performance. The Kappa under KNN varies

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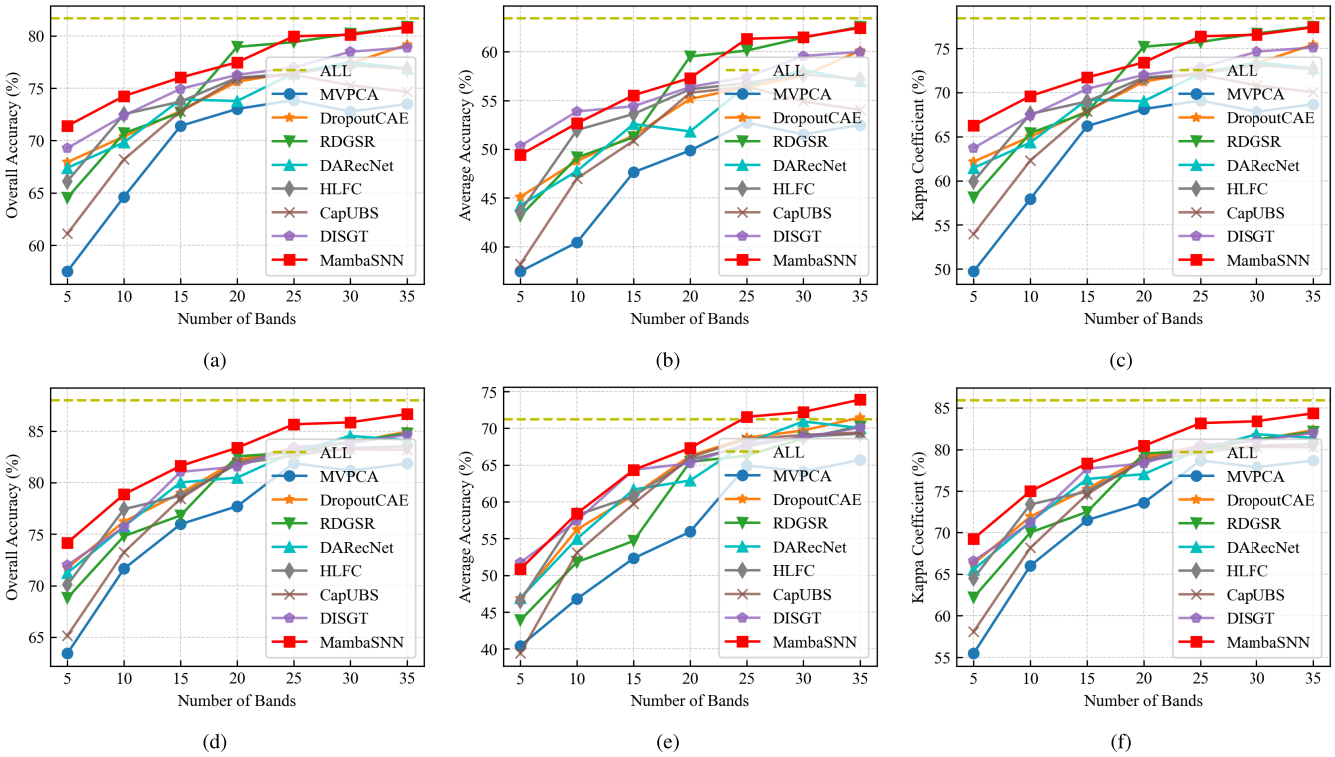


Fig. 7. Classification performance of different BS methods under different band subset sizes on HC dataset. (a)–(c) KNN classification OA, AA, and Kappa results based on different BS methods for the selected bands. (d)–(f) SVM classification OA, AA, and Kappa results.

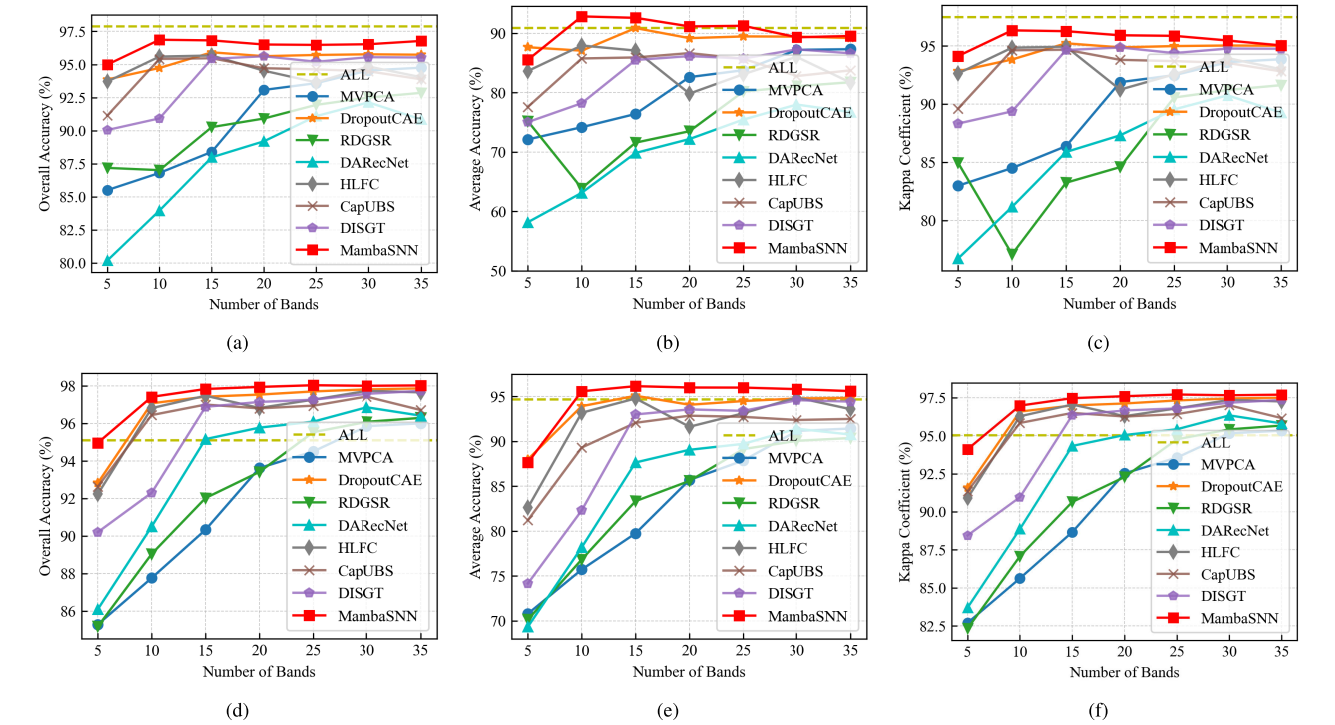


Fig. 8. Classification performance of different BS methods under different band subset sizes on TLS dataset. (a)–(c) KNN classification OA, AA, and Kappa results based on different BS methods for the selected bands. (d)–(f) SVM classification OA, AA, and Kappa results.

by about 10.15%, which is smaller than the fluctuations observed in DISGT, HLFC, and DropoutCAE. It indicates that MambaSNN mitigates the Hughes phenomenon and maintains relatively consistent accuracy as the number of selected bands increases.

Fig. 6 presents the results on the H dataset, highlighting MambaSNN's superiority in classification accuracy beyond the 10-band configuration under both classifiers. Under KNN classification with 35 selected bands, MambaSNN achieves an OA of 87.31%, marginally outperforming HLFC by 0.83% and

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TABLE III
SVM CLASSIFICATION ACCURACY FOR 15 SELECTED BANDS ON THE PU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	MVPCA	DARecNet	RDGSR	DropoutCAE	HLFC	CapUBS	DISGT	MambaSNN
Asphalt	91.25±2.69	91.93±0.50	89.63±1.04	89.24±0.80	**92.76±0.51**	90.07±0.85	91.16±0.90	91.91±0.78
Meadows	95.45±0.27	97.10±0.28	97.48±0.32	97.26±0.22	97.14±0.25	96.93±0.31	97.66±0.20	**97.71±0.21**
Gravel	52.12±2.73	69.41±1.44	70.61±1.36	68.99±1.30	76.35±1.71	70.79±2.06	71.20±1.87	**77.27±1.39**
Trees	73.74±1.50	93.22±0.86	**93.44±1.19**	92.33±0.87	89.94±0.80	92.56±0.78	92.92±0.83	93.33±0.77
Painted Metal Sheets	98.45±0.47	99.39±0.28	99.25±0.27	99.21±0.36	99.13±0.37	99.27±0.33	**99.48±0.34**	99.36±0.27
Bare Soil	71.39±0.94	82.13±1.00	82.36±1.15	84.28±0.99	83.42±0.82	78.13±1.30	84.02±1.06	**84.43±0.77**
Bitumen	13.41±8.10	79.16±1.89	71.50±3.65	72.55±3.23	**82.11±2.65**	76.70±2.41	76.41±2.79	77.88±3.50
Self-Blocking Bricks	81.63±1.88	87.13±1.48	87.00±1.24	85.96±1.35	88.41±1.37	86.26±1.16	87.33±1.50	**89.32±1.30**
Shadows	**99.96±0.05**	99.72±0.15	99.80±0.15	99.82±0.17	99.74±0.15	99.73±0.17	99.92±0.17	99.78±0.14
OA(%)	84.74±0.22	91.62±0.11	91.27±0.12	91.13±0.19	92.22±0.11	90.64±0.15	92.29±0.14	**92.69±0.06**
AA(%)	75.27±1.67	88.80±0.35	87.90±0.39	87.74±0.42	89.90±0.27	87.81±0.24	89.14±0.31	**90.11±0.36**
Kappa(%)	79.40±0.32	88.83±0.16	88.36±0.15	88.19±0.25	89.62±0.14	87.51±0.20	89.73±0.18	**90.26±0.08**

DISGT by 0.49%. This advantage becomes more pronounced under SVM classification. To reach an OA of 90%, MambaSNN requires only 15 bands, whereas other methods need 20 bands to attain the same performance. When the selected bands increase to 25, MambaSNN further boosts performance to 91.54% OA, compared to 90.92% for HLFC and 90.85% for DISGT. As the number of selected bands exceeds 25, nearly all methods except MVPCA show convergence in performance. Notably, at 35 bands under SVM classification, MambaSNN attains an OA of 92.07%, approaching the full-band result of 92.70% while utilizing only 17% of the original spectral channels. These findings demonstrate MambaSNN's remarkable spectral effectiveness in high-dimensional HSI scenarios.

As shown in Fig. 7, MambaSNN also exhibits superior performance on the HC dataset across different band subset cardinalities. Under KNN classification, MambaSNN consistently outperforms competing approaches. At the 15-band configuration, it achieves 76.02% OA and 55.54% AA, representing improvements of 1.08% in OA and 1.13% in AA over DISGT, and surpassing CapUBS by 3.23% and 4.66%, respectively. This advantage becomes more pronounced as the number of selected bands increases. With 25 bands, MambaSNN attains 79.95% OA and 61.33% AA, outperforming DISGT by 2.97% and 3.93%, respectively. Under SVM classification, MambaSNN maintains clear superiority. At 15 bands, it reaches 81.63% OA and 64.32% AA, with a Kappa (78.32%) higher than all competing methods. At 25 bands, MambaSNN achieves 85.67% OA and 71.54% AA, improving upon DISGT by 2.23% and 3.61%, and outperforming CapUBS by 2.56% and 3.05%. Notably, at 35 bands, MambaSNN further boosts performance to 86.66% OA and 73.88% AA, surpassing the best competing baseline (DropoutCAE) by 1.72% OA and 2.44% AA.

MambaSNN consistently outperforms competing methods across different band subset cardinalities on the TLS dataset (see Fig. 8). Under KNN classification, MambaSNN achieves the highest accuracy across all scenarios, reaching 96.87% OA and 92.85% AA at 10 bands. These results notably exceed HLFC by significant margins in both metrics. Even as the number of selected bands increases, MambaSNN maintains stable and superior performance, with the Kappas remaining above 95%, indicating strong robustness to varying spectral constraints. Under SVM classification, the advantage of MambaSNN becomes more pronounced. At 15 bands, it attains 97.84% OA and 96.15% AA, outperforming the second-best method (DISGT) by 0.15% in OA and 2.15% in AA. As the selected bands increase to 25–35, MambaSNN further improves to around 98% OA and 95.80% AA. This progression is consistent with the observed superiority of MambaSNN over CapUBS, HLFC, and DropoutCAE. The corresponding Kappas reach up to 97.71%, confirming its strong discriminative capability.

To ensure a rigorous and equitable basis for quantitative comparison across methods, convergence thresholds are determined empirically for each dataset. These thresholds should correspond to inflection points beyond which the inclusion of additional spectral bands leads to marginal performance improvements, indicating asymptotic stability. This selection can mitigate the risk of the Hughes phenomenon commonly associated with suboptimal BS. Furthermore, this strategy can facilitate consistent cross-method evaluation. According to the aforementioned criteria, the band thresholds for the PU, IP, H, HC, and TLS datasets are 15, 20, 20, 25, and 20, respectively. Tables III–VII present detailed SVM classification metrics at the respective convergence thresholds for each dataset.

Table III shows the SVM classification performance for 15 selected bands on the PU dataset based on different BS methods. From the per-class classification accuracy, the proposed MambaSNN achieves the highest classification accuracy in four categories, including meadows (97.71%), gravel (77.27%), bare soil (84.43%), and self-blocking bricks (89.32%). Besides, it remains virtually tied for the top score in painted metal sheets and shadows, differing by only 0.03% and 0.18%, respectively. At the aggregate level, MambaSNN attains the highest OA (92.69%), AA (90.11%), and Kappa (90.26%), consistently outperforming established baselines such as HLFC, DARecNet, and RDGSR.

Table IV shows the SVM classification performance for 20 selected bands on the IP dataset across different BS methods. The proposed MambaSNN achieves the highest classification accuracy in 11 categories, including hay-windrowed (97.56%), woods (93.03%), stone-steel-towers (85.48%), and so on. MambaSNN performs particularly well in alfalfa, corn-notill, and corn categories, improving the classification accuracy by 15.85%, 3.97%, and 5.96%, respectively, compared to the best results from other methods. Furthermore, MambaSNN outperforms all other comparison methods in OA, AA, and Kappa. When compared to the second-best method of DARecNet,

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TABLE IV

SVM CLASSIFICATION ACCURACY FOR 20 SELECTED BANDS ON THE IP DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	MVPCA	DARecNet	RDGSR	DropoutCAE	HLFC	CapUBS	DISGT	MambaSNN
Alfalfa	19.27±17.19	50.73±17.75	28.05±11.96	56.10±15.23	44.63±11.44	61.22±12.07	**75.61±10.80**	71.95±11.86
Corn-notill	46.89±4.20	71.28±2.56	61.20±3.38	71.35±1.95	66.71±3.30	66.40±2.28	70.75±1.89	**75.32±1.83**
Corn-mintill	37.40±2.58	67.00±3.40	47.43±3.01	63.99±3.13	62.99±3.12	58.71±2.81	63.56±3.59	**68.26±2.47**
Corn	43.29±8.88	53.76±6.54	40.47±5.92	51.69±6.09	46.43±3.63	42.63±4.58	53.29±5.94	**59.72±6.51**
Grass-pasture	82.14±2.80	85.45±4.61	86.21±1.99	84.51±2.59	83.06±4.25	87.77±3.29	**91.52±2.91**	86.76±2.61
Grass-trees	92.62±3.01	91.46±3.44	90.91±3.01	89.95±2.86	**92.89±3.45**	87.90±2.39	92.28±2.22	91.70±2.31
Grass-pasture-mowed	52.80±10.24	**83.60±6.31**	75.60±7.89	70.80±17.07	49.60±12.92	36.80±18.31	66.00±19.78	78.40±5.71
Hay-windrowed	96.44±2.38	96.60±1.04	91.47±3.36	96.53±2.16	93.63±3.13	96.19±1.38	97.54±0.62	**97.56±1.04**
Oats	8.89±10.89	41.67±12.97	17.22±10.08	36.11±15.96	31.11±12.72	20.56±11.12	**43.33±15.87**	37.78±10.77
Soybean-notill	33.75±5.57	**75.76±1.35**	66.63±2.94	72.23±1.59	66.29±2.32	67.69±3.07	71.30±2.64	75.37±2.17
Soybean-mintill	76.29±2.86	80.05±1.67	77.14±1.75	77.81±2.22	74.76±2.49	74.85±1.99	80.30±1.44	**80.38±1.69**
Soybean-clean	28.93±5.37	66.14±4.16	42.42±4.61	60.92±3.71	50.32±4.28	47.72±2.69	66.76±3.61	**69.61±3.42**
Wheat	91.46±2.89	92.05±4.55	94.86±2.86	**95.62±2.51**	90.97±3.40	92.27±2.86	95.35±3.26	93.08±3.09
Woods	92.27±1.76	92.40±1.40	92.33±1.87	91.87±2.09	89.47±2.11	92.00±2.21	**95.45±1.18**	93.03±1.44
Buildings-Grass-Trees-Drives	40.98±4.24	44.50±5.97	39.63±4.71	42.68±3.78	45.36±4.35	44.50±4.61	**51.30±4.21**	48.85±5.47
Stone-Steel-Towers	74.05±7.96	79.88±7.94	80.48±6.61	81.90±7.98	76.43±5.16	79.29±6.48	81.31±8.42	**85.48±4.79**
OA(%)	64.36±0.75	78.02±0.34	71.25±0.40	76.35±0.86	73.14±0.89	73.00±0.54	78.55±0.33	**79.58±0.48**
AA(%)	57.34±1.76	73.27±1.19	64.50±1.19	71.50±2.40	66.54±1.91	66.03±1.82	74.74±1.87	**75.83±1.28**
Kappa(%)	58.63±0.96	74.86±0.38	67.05±0.43	72.94±0.98	69.29±0.99	69.13±0.61	75.43±0.39	**76.66±0.55**

TABLE V

SVM CLASSIFICATION ACCURACY FOR 20 SELECTED BANDS ON THE H DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	MVPCA	DARecNet	RDGSR	DropoutCAE	HLFC	CapUBS	DISGT	MambaSNN
Grass Healthy	95.79±4.14	97.95±0.57	**98.30±0.47**	97.72±0.64	97.54±1.25	97.81±0.82	97.54±1.04	97.86±1.04
Grass Stressed	88.24±5.10	97.49±0.90	97.28±1.08	97.69±0.79	97.78±1.02	97.77±0.80	97.11±1.20	**97.84±1.00**
Grass Synthetic	98.76±0.48	99.39±0.20	99.62±0.24	99.52±0.23	99.60±0.22	99.57±0.23	**99.67±0.20**	99.57±0.20
Tree	92.17±1.84	**98.03±0.65**	95.43±1.62	96.86±0.90	97.47±1.25	97.23±1.03	95.93±1.84	96.79±1.30
Soil	95.14±1.58	98.37±0.64	98.40±0.43	**98.49±0.42**	98.25±0.75	98.48±0.48	98.25±0.53	98.33±0.70
Water	87.02±3.77	94.49±3.00	95.14±3.28	91.78±3.76	95.07±2.95	95.14±3.11	94.93±4.03	**96.23±2.22**
Residential	79.67±2.63	88.90±1.80	88.26±1.48	89.02±1.74	87.47±1.11	87.90±2.07	88.12±1.80	**89.20±1.69**
Commercial	61.45±3.62	84.92±1.65	**85.66±1.31**	82.62±1.61	84.08±1.26	82.92±1.68	83.00±2.16	85.15±1.32
Road	77.17±2.49	84.98±1.61	85.03±2.05	81.90±2.43	84.18±2.06	**85.05±1.66**	83.71±2.33	83.06±2.19
Highway	67.59±6.52	89.49±2.42	89.43±1.51	90.87±1.73	90.98±1.77	89.70±1.83	89.76±2.24	**91.54±1.78**
Railway	62.28±3.17	85.63±1.78	87.70±1.56	**88.71±1.59**	86.18±2.13	83.58±2.65	87.04±2.00	87.21±2.04
Parking Lot1	66.21±3.41	84.19±1.76	80.33±1.57	81.86±2.25	80.23±1.83	83.98±1.19	81.21±2.53	**84.78±2.20**
Parking Lot2	10.43±2.05	36.92±5.96	37.20±5.64	34.41±6.67	36.40±5.88	36.07±4.38	38.46±4.32	**40.21±6.54**
Tennis Court	98.16±1.09	98.73±1.05	98.47±0.77	98.44±1.28	99.12±0.94	98.70±0.97	98.42±1.00	**99.35±0.88**
Running Track	98.77±0.48	98.94±0.49	98.87±0.61	**98.99±0.43**	98.82±0.56	98.87±0.48	98.80±0.45	98.94±0.69
OA(%)	79.05±0.78	90.35±0.29	90.03±0.27	89.86±0.41	89.88±0.26	89.88±0.39	90.16±0.20	**90.66±0.32**
AA(%)	78.59±0.73	89.23±0.42	89.01±0.44	88.59±0.59	88.88±0.46	88.85±0.29	89.28±0.35	**89.74±0.44**
Kappa(%)	77.32±0.84	89.56±0.31	89.22±0.29	89.03±0.45	89.05±0.28	89.05±0.42	89.35±0.21	**89.90±0.35**

TABLE VI

SVM CLASSIFICATION ACCURACY FOR 25 SELECTED BANDS ON THE HC DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	MVPCA	DARecNet	RDGSR	DropoutCAE	HLFC	CapUBS	DISGT	MambaSNN
Strawberry	92.40±0.09	93.61±0.25	94.90±0.17	94.19±0.15	93.09±0.17	93.91±0.19	94.44±0.34	**95.24±0.22**
Cowpea	77.33±0.64	73.91±0.32	71.79±0.57	71.50±0.25	73.59±0.84	72.79±0.24	72.86±0.69	**79.43±0.37**
Soybean	72.90±0.89	74.26±1.30	73.07±1.04	72.18±0.80	75.82±0.82	73.28±0.69	68.91±0.74	**76.01±0.41**
Sorghum	82.63±0.32	90.62±0.62	87.15±0.77	90.24±0.65	89.27±0.74	90.62±0.65	**92.73±0.41**	92.37±0.36
Water spinach	17.81±2.05	28.22±2.18	28.17±1.20	29.54±2.78	26.09±3.56	28.15±2.97	26.78±2.13	**36.30±1.28**
Watermelon	19.03±0.87	24.74±0.76	4.71±0.51	24.45±1.11	22.22±1.33	**26.81±0.48**	15.14±0.69	22.49±0.81
Greens	71.38±1.04	76.41±0.75	76.88±1.43	77.30±0.92	77.08±1.73	76.84±1.07	**78.60±1.87**	78.56±1.07
Trees	69.80±0.66	67.60±0.55	66.11±0.95	66.88±0.77	69.94±0.73	69.64±0.34	71.06±0.33	**75.11±0.82**
Grass	65.97±0.89	60.96±0.54	51.88±0.57	58.37±0.73	59.76±0.56	62.57±0.52	61.31±0.70	**69.92±0.54**
Red roof	75.80±1.03	85.37±0.35	**92.96±0.15**	91.52±0.33	80.91±0.67	89.19±0.23	92.42±0.25	91.37±0.26
Gray roof	75.64±1.05	83.17±0.32	90.12±0.24	**88.16±0.40**	80.37±0.73	81.46±0.37	87.34±0.37	86.99±0.81
Plastic	28.18±0.71	29.74±1.69	29.54±0.82	36.09±1.77	35.14±1.38	31.14±2.15	30.72±1.15	**36.46±1.02**
Bare soil	50.38±0.98	51.75±1.16	46.52±1.09	51.39±1.14	53.18±0.83	51.73±0.51	49.90±0.50	**54.71±1.13**
Road	79.11±0.51	80.65±0.60	82.74±0.30	80.15±0.27	81.12±0.55	80.79±0.42	80.44±0.34	**83.48±0.61**
Bright object	65.32±1.20	70.23±1.42	68.94±0.63	68.53±2.24	68.28±2.66	70.88±1.48	69.51±0.87	**71.31±1.20**
Water	98.27±0.12	97.94±0.06	**98.54±0.04**	98.18±0.07	97.69±0.12	97.66±0.06	98.04±0.07	98.22±0.12
OA(%)	81.87±0.10	82.92±0.06	82.96±0.04	83.28±0.03	82.63±0.07	83.08±0.05	83.45±0.10	**85.68±0.07**
AA(%)	65.12±0.18	68.07±0.18	66.50±0.17	68.67±0.19	67.72±0.33	68.59±0.26	68.14±0.28	**71.75±0.11**
Kappa(%)	78.66±0.12	79.95±0.08	79.97±0.05	80.37±0.04	79.59±0.08	80.13±0.06	80.57±0.12	**83.20±0.08**

MambaSNN improves OA, AA, and Kappa by 1.56%, 2.56%, and 1.80%, respectively.

On the H dataset with 20 selected bands, the proposed MambaSNN achieves the highest classification accuracy in

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TABLE VII
SVM CLASSIFICATION ACCURACY FOR 20 SELECTED BANDS ON THE TLS DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	MVPCA	DARecNet	RDGSR	DropoutCAE	HLFC	CapUBS	DISGT	MambaSNN
Orange tile	99.37±0.10	99.60±0.10	99.61±0.17	99.78±0.13	99.75±0.07	99.76±0.12	99.68±0.08	**99.84±0.10**
Slate	48.13±1.41	84.95±1.41	76.87±1.76	91.83±1.14	88.79±0.65	90.38±1.02	88.49±0.61	**92.46±0.81**
Dark fiber cement	99.46±0.18	99.79±0.05	99.15±0.26	**99.83±0.05**	99.82±0.03	99.82±0.03	99.83±0.09	99.82±0.06
Dark sheet metal	78.94±1.38	90.10±1.45	83.82±1.55	94.59±1.03	93.59±0.54	94.32±0.64	91.46±0.58	**94.63±0.69**
Dark asphalt	94.71±1.27	94.52±0.63	91.53±1.03	96.36±0.65	96.13±0.42	96.27±0.44	96.65±1.13	**96.89±0.53**
Red asphalt	85.25±1.39	91.92±1.40	91.90±0.96	95.00±1.89	96.79±1.65	95.68±1.46	97.82±0.56	**98.24±0.56**
Green-painted asphalt	22.45±7.19	62.86±8.50	21.63±7.81	71.84±12.27	67.76±8.20	78.37±5.26	82.45±8.72	**89.39±3.51**
White road marking	85.86±3.24	68.80±4.87	68.80±6.25	86.49±4.57	77.49±6.00	80.31±4.31	79.37±6.43	**88.69±2.76**
Brown paving stone	89.33±3.47	89.47±1.23	82.86±1.18	92.13±1.83	94.13±1.40	91.52±2.40	93.09±2.83	**94.45±1.95**
Clear paving stone	96.71±1.35	91.27±1.72	91.39±1.94	97.34±0.63	97.06±0.77	95.48±1.58	98.14±0.38	**98.43±0.63**
Healthy grass	91.01±1.00	90.03±1.14	75.47±1.23	92.86±0.99	92.99±0.79	92.27±0.72	91.17±0.69	**93.01±1.23**
Tree	**99.83±0.07**	99.39±0.13	96.55±0.21	99.61±0.11	99.56±0.14	99.40±0.19	99.28±0.16	99.43±0.15
Bare soil	98.10±0.42	93.78±1.53	93.68±1.08	97.93±0.58	97.63±0.44	97.86±0.65	97.86±0.64	**98.39±0.91**
Gravels	98.22±0.25	97.43±0.40	97.37±0.97	99.05±0.39	99.00±0.33	98.29±0.26	99.25±0.28	**99.44±0.32**
OA(%)	93.67±0.16	95.79±0.13	92.96±0.06	97.58±0.14	97.40±0.09	97.24±0.15	97.28±0.08	**97.98±0.11**
AA(%)	84.81±0.57	89.56±0.87	83.62±0.83	93.90±1.15	92.89±0.93	93.55±0.58	93.90±1.05	**95.94±0.35**
Kappa(%)	92.57±0.19	95.07±0.16	91.74±0.07	97.16±0.16	96.95±0.11	96.76±0.17	96.82±0.09	**97.63±0.12**

seven categories (see Table V). Specifically, it performs best in natural element categories such as water (96.23%) and grass stressed (97.84%). This demonstrates its sensitivity to subtle spectral variations. Moreover, MambaSNN ranks first in several artificial land cover classes, including residential (89.20%), highway (91.54%), and tennis court (99.35%). It also outperforms other methods in the challenging parking lot1 (84.78%) and parking lot2 (40.21%) categories, significantly improving classification performance in low-resolution scenarios. Besides, the difference between MambaSNN and the best results in most nonoptimal categories is less than 1% (e.g., Grass Synthetic only differs by 0.05%). Notably, MambaSNN outperforms all other comparison methods in OA, AA, and Kappa.

Table VI shows the SVM classification performance for 25 selected bands on the HC dataset. MambaSNN achieves the highest classification accuracy in ten categories, including strawberry (95.24%), road (83.48%), and cowpea (79.43%). Specifically, it has demonstrated proficiency in challenging categories, including water spinach and trees, exhibiting a 7.77% and 4.05% improvement in classification accuracy over the second-best method, respectively. At the aggregate level, MambaSNN achieves 85.68% OA, 71.75% AA, and 83.20% Kappa, outperforming the second-best method (DISGT) by 2.23% in OA, 3.61% in AA, and 2.63% in Kappa. These results highlight MambaSNN's superior performance across different categories and spectral constraints.

As shown in Table VII, MambaSNN attains the highest classification accuracy in 12 categories on the TLS dataset, including challenging surfaces such as slate (92.46%), dark sheet metal (94.63%), red asphalt (98.24%), and green-painted asphalt (89.39%). Furthermore, it is demonstrated to be superior in texture-sensitive classes such as white road marking and clear paving stone, where other methods exhibit substantial performance degradation. In the remaining categories, its performance is nearly tied with the best results, typically differing by less than 0.3%. At the overall level, MambaSNN achieves the highest OA (97.98%), AA (95.94%), and Kappa (97.63%), consistently outperforming all other baselines.

To further illustrate the differences between the band subsets of different methods, Fig. 9 presents the distribution of selected bands across all datasets. The visualization reveals that MambaSNN consistently achieves more uniform spectral coverage compared to competing approaches, which tend to exhibit clustering in specific wavelength regions. Such uniform coverage is advantageous for capturing diverse material signatures and avoiding spectral bias, which may explain the superior classification performance observed across heterogeneous land cover types.

To intuitively evaluate spatial consistency and classification accuracy, we visualize the SVM classification results produced from the selected bands of all comparative methods on the PU, IP, H, HC, and TLS datasets (see Figs. 10–14).

For the PU dataset (see Fig. 10), our method exhibits superior overall classification performance, producing cleaner and more spatially coherent maps. In particular, it demonstrates better discrimination in challenging regions such as gravel and bare soil, where other methods tend to show class confusion and noisy boundaries. On the IP dataset (see Fig. 11), our approach produces a more homogeneous and accurate classification result compared with other methods. Notably, it shows improved identification in spectrally similar or highly mixed classes such as alfalfa and corn-mintill in the red box, which are often fragmented or misclassified by competing approaches. For the H dataset (see Fig. 12), the proposed method achieves a high degree of spatial consistency while preserving detailed urban structures. In zoomed regions such as parking areas and railway tracks, clearer interclass boundaries and fewer misclassifications can be observed, indicating the robustness of the selected bands in structurally complex scenes. In the case of the HC dataset (see Fig. 13), the classification results produced by MambaSNN show superior spatial consistency and accuracy compared with other methods. In categories such as soybean and trees, our method effectively preserves spatial structures and generates cleaner classification maps, reflecting its ability to capture fine spectral–spatial details. For the TLS dataset (see Fig. 14), MambaSNN produces the most coherent and noise-free classification maps, especially in regions with complex materials

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TABLE VIII
RUNTIME (S) AND OA (%) OF DIFFERENT BS METHODS ON FIVE DATASETS

Methods	PU (15)		IP (20)		H (20)		HC (25)		TLS (20)	
	Time (s)	OA (%)	Time (s)	OA (%)	Time (s)	OA (%)	Time (s)	OA (%)	Time (s)	OA (%)
MVPCA	1.60	84.74	0.39	64.36	7.74	79.05	8.32	81.87	35.07	93.67
RDGSR	90.79	91.27	54.31	71.25	49.31	90.03	157.32	82.96	122.79	92.96
HLFC	214.56	92.22	254.54	73.14	393.66	89.88	328.26	82.63	1076.05	97.40
DARecNet	609.89	91.62	279.46	78.02	568.54	90.35	7339.83	82.92	2677.41	95.79
DropoutCAE	618.95	91.13	170.68	76.35	240.97	89.86	1973.65	83.28	1725.92	97.58
CapUBS	954.56	90.64	522.14	73.00	758.26	89.88	6813.25	83.08	2367.45	97.24
DISGT	539.82	92.29	246.36	78.55	486.23	90.16	5832.15	83.45	2666.98	97.28
MambaSNN	**252.99**	**92.69**	**125.43**	**79.58**	**209.58**	**90.66**	**1243.82**	**85.68**	**1620.31**	**97.98**

Note: The numbers in parentheses following each dataset name indicate the corresponding number of selected spectral bands.

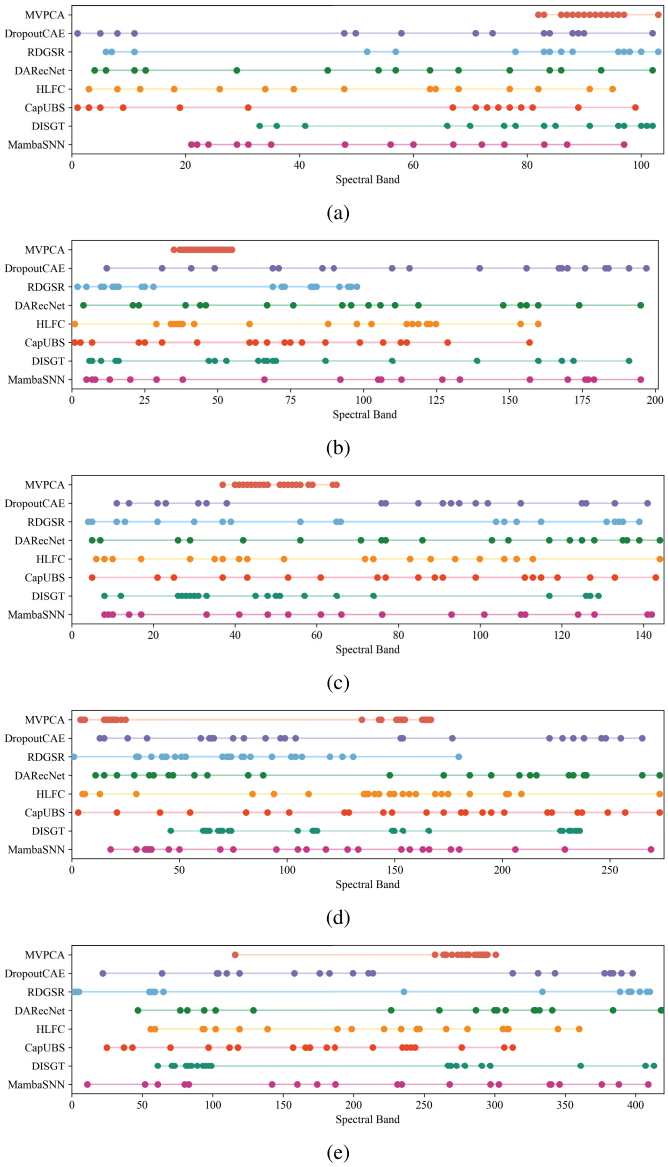


Fig. 9. Distribution of spectral bands selected based on different BS methods. (a) PU (15 bands), (b) IP (20 bands), (c) H (20 bands), (d) HC (25 bands), and (e) TLS (20 bands).

such as asphalt, sheet metal, and paving stones. Its selected 20-band subset preserves fine structural details and yields clearer class boundaries than all competing methods.

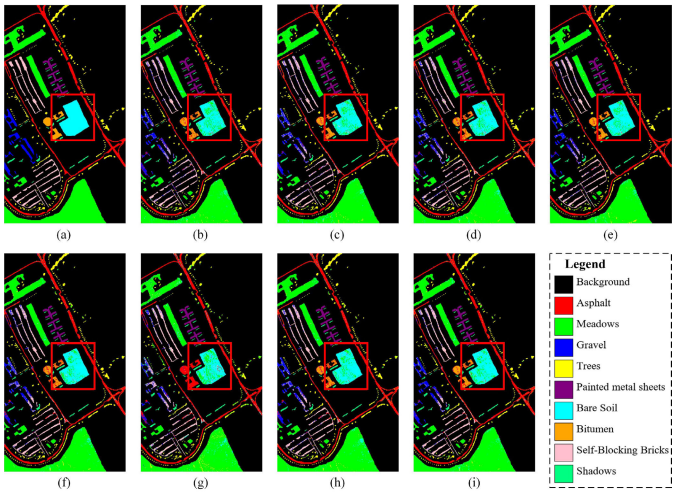


Fig. 10. SVM classification maps for different methods on the PU dataset. (a) Ground-truth. (b) CapUBS. (c) DARecNet. (d) DropoutCAE. (e) HLFC. (f) DISGT. (g) MVPCA. (h) RDGSR. (i) MambaSNN.

These improvements can be attributed to the design of the proposed framework. The SpeMamba module effectively models long-range spectral dependencies through state-space sequence modeling, enabling the extraction of more discriminative spectral features. Meanwhile, the SNNRecNet module promotes sparse and noise-resilient feature reconstruction, which helps reduce spectral redundancy and suppress noisy responses. The combination of these two components leads to more stable BS and ultimately produces more accurate and spatially consistent classification results. These qualitative observations corroborate the quantitative comparisons presented in Tables III–VII, underscoring MambaSNN's ability to produce band subsets that are simultaneously spectrally discriminative and spatially coherent.

D. Analysis of Computational Efficiency

To assess the computational efficiency of different BS algorithms, we conduct a comprehensive runtime performance comparison across the five benchmark datasets (see Table VIII). It demonstrates that MambaSNN achieves substantial performance efficiency over other deep learning baselines such as DropoutCAE and CapUBS. This acceleration is attributed to the near-linear computational complexity of its SSM, which effectively circumvents computational

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TABLE IX

COMPARISON OF GPU MEMORY CONSUMPTION (MB) AND NUMBER OF TRAINABLE PARAMETERS (M) OF DIFFERENT DEEP LEARNING-BASED BS METHODS ACROSS FIVE DATASETS

Methods	PU		IP		H		HC		TLS	
	Mem(MB)	Param(M)	Mem(MB)	Param(M)	Mem(MB)	Param(M)	Mem(MB)	Param(M)	Mem(MB)	Param(M)
DARecNet	666	0.16	1168	0.20	886	0.18	2074	0.24	2236	0.37
DropoutCAE	822	**0.015**	1827	**0.029**	835	**0.023**	1524	**0.040**	2448	**0.059**
CapUBS	3334	299.38	8448	1190.81	2964	598.76	20506	2323.88	20062	3114.83
DISGT	804	0.68	1626	1.16	2872	0.75	7530	1.33	9388	2.29
MambaSNN	**558**	0.45	**912**	0.67	**642**	0.51	**1770**	0.92	**1868**	1.42

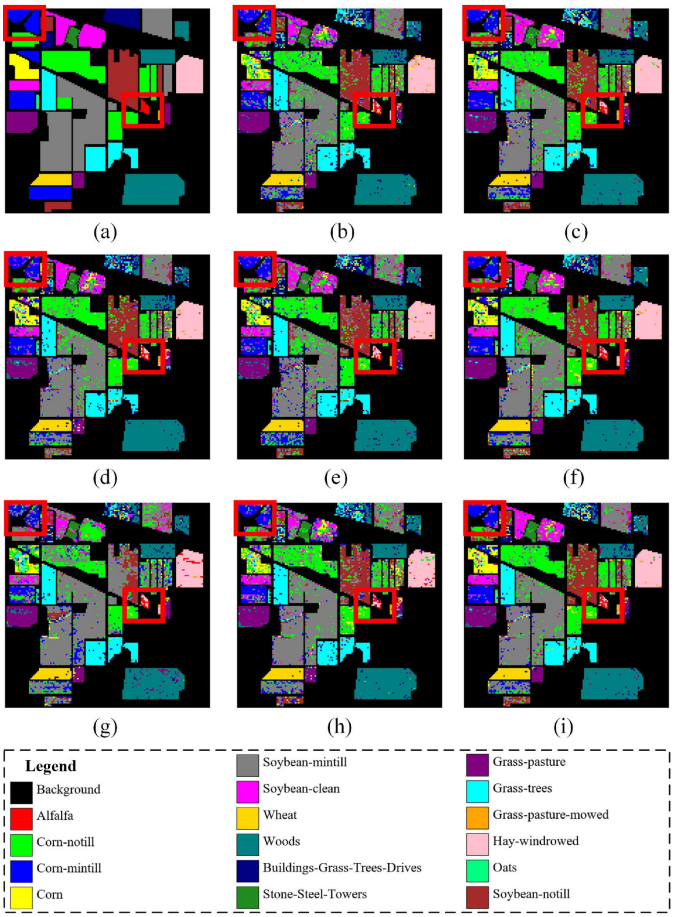


Fig. 11. SVM classification maps for different methods on the IP dataset. (a) Ground-truth. (b) CapUBS. (c) DARecNet. (d) DropoutCAE. (e) HLFC. (f) DISGT. (g) MVPCA. (h) RDGSR. (i) MambaSNN.

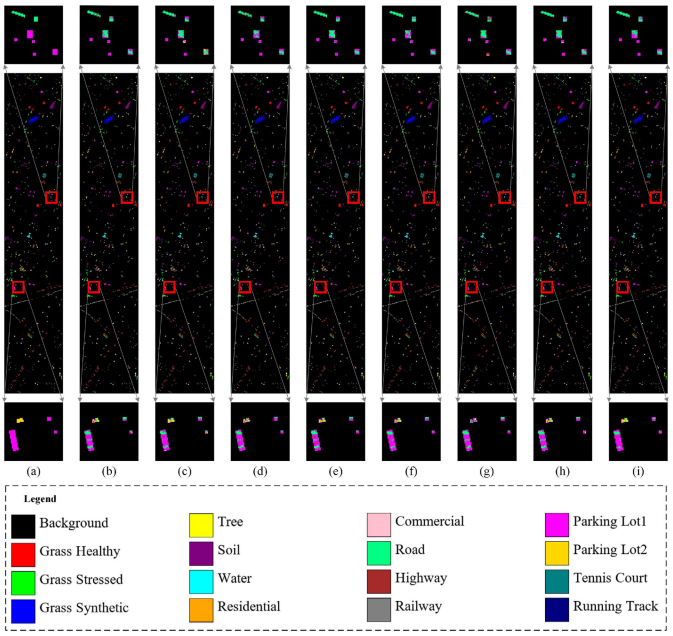


Fig. 12. SVM classification maps for different methods on the H dataset. (a) Ground-truth. (b) CapUBS. (c) DARecNet. (d) DropoutCAE. (e) HLFC. (f) DISGT. (g) MVPCA. (h) RDGSR. (i) MambaSNN.

redundancies inherent in convolutional operations. Although the iterative optimization and BP processes of MambaSNN impose greater computational overhead than algebraic approaches (e.g., MVPCA and RDGSR), these costs are offset by significant classification gains. In addition, compared with the relatively complex traditional method HLFC, MambaSNN also exhibits higher efficiency on the IP and H datasets, while simultaneously achieving higher OA. For instance, when selecting 20 bands from the IP dataset, MambaSNN elevates OA by 6.44% compared to HLFC. Concurrently, the runtime is reduced from 254.54 to 125.43 s by MambaSNN, marking a 50.72% decrease. The end-to-end task-driven optimization ensures that the learned band subsets are closely aligned with the downstream classification objective, thereby directly contributing to these performance gains.

Following the runtime analysis, Table IX further presents GPU memory consumption and the number of trainable parameters for each deep learning-based method. MambaSNN consistently exhibits the lowest memory footprint across all five datasets, requiring only 558, 912, 642, 1770, and 1868 MB on the PU, IP, H, HC, and TLS datasets, respectively. Relative to the second memory-efficient methods, this represents 11%–32% reductions on PU, IP, and H, with the advantage becoming even more pronounced on the large-scale datasets HC and TLS. Importantly, this memory efficiency is achieved without employing an over-parameterized design. MambaSNN maintains a compact model architecture with 0.45–1.42 million parameters across datasets, which is comparable to other lightweight baselines (e.g., DARecNet and DISGT) and substantially smaller than the heavy-capacity CapUBS (ranging from 299 to 3115 million parameters). This indicates that MambaSNN delivers strong scalability primarily through an efficient architectural design rather than parameter inflation.

When evaluated in conjunction with the runtime results (see Table VIII), the comprehensive memory-parameter-runtime

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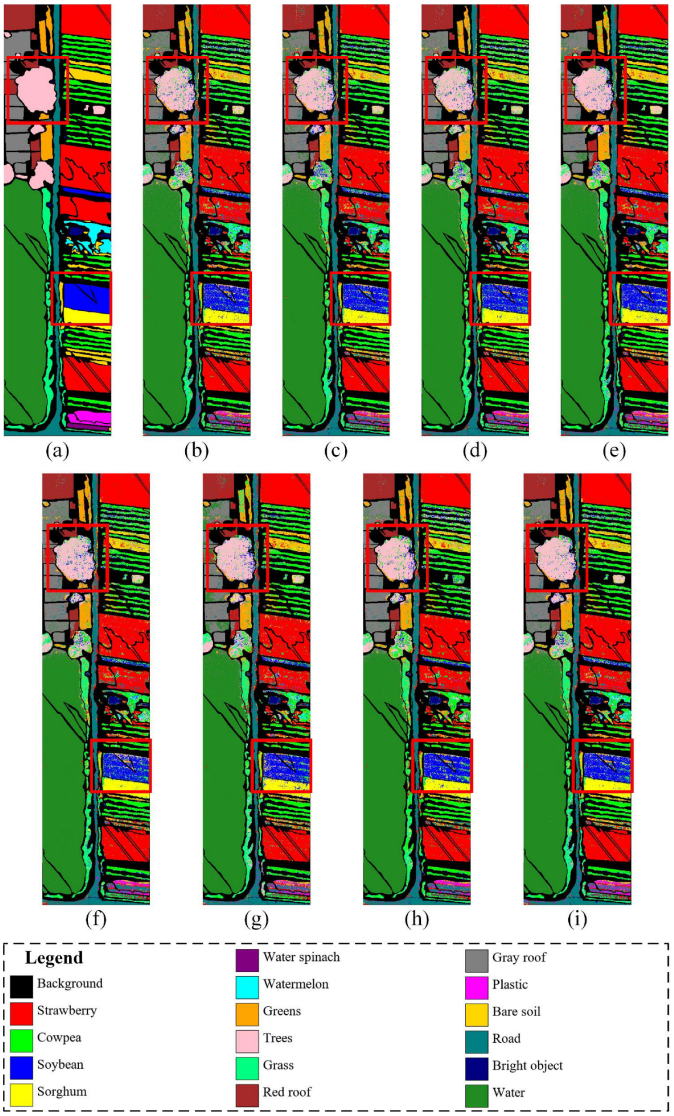


Fig. 13. SVM classification maps for different methods on the HC dataset. (a) Ground-truth. (b) CapUBS. (c) DARecNet. (d) DropoutCAE. (e) HLFC. (f) DISGT. (g) MVPCA. (h) RDGSR. (i) MambaSNN.

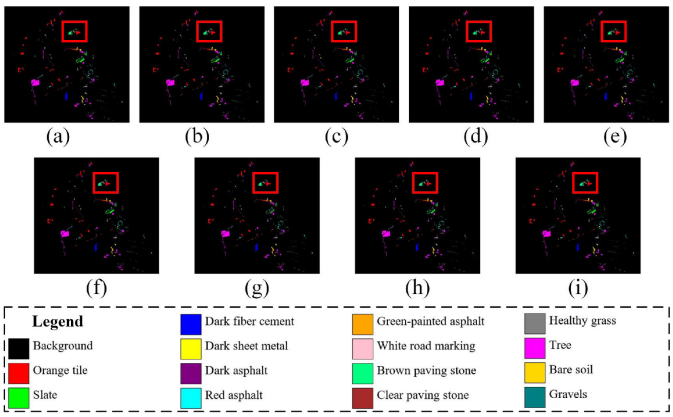


Fig. 14. SVM classification maps for different methods on the TLS dataset. (a) Ground-truth. (b) CapUBS. (c) DARecNet. (d) DropoutCAE. (e) HLFC. (f) DISGT. (g) MVPCA. (h) RDGSR. (i) MambaSNN.

profile substantiates that MambaSNN provides a highly advantageous efficiency tradeoff. The proposed method exhibits the minimal GPU memory requirements and compact parameter budgets. Notably, it circumvents the excessive computational and storage overhead observed in large-scale models such as CapUBS. These properties make MambaSNN particularly well-suited for resource-constrained deployment scenarios and large-scale or near-real-time hyperspectral BS applications.

TABLE X

ABLATION ANALYSIS FOR MOA USING THE SVM CLASSIFIER ON FIVE DATASETS (%). THE HIGHEST SCORES ARE HIGHLIGHTED IN BOLD

Methods	PU	IP	H	HC	TLS
MambaSNN	**91.40**	**75.15**	**89.90**	**82.33**	**97.49**
MambaCNN	91.05	74.94	88.16	81.91	97.08
Without SNN	90.82	74.75	87.84	81.72	96.86
Without Mamba	88.71	74.37	87.25	80.15	96.59

E. Ablation Study

To evaluate the contribution of each core module, a systematic ablation experiment was carried out under SVM classification. In this experiment, two ablated scenarios, including removing the Mamba backbone ("Without Mamba") and omitting the SNN reconstruction module ("Without SNN"), were tested. Meanwhile, to further validate the effect of the SNN decoder, we constructed a third scenario that replaces the SNN decoder with a conventional CNN-based reconstruction module, designated as "MambaCNN."

The results are presented in Table X, which reports the mean OA (mOA) computed across band subsets ranging from 5 to 35 spectral channels for each experimental scenario. The findings reveal a consistent performance hierarchy across all benchmark datasets (MambaSNN > MambaCNN > "Without SNN" > "Without Mamba"), providing empirical evidence for the complementary nature of both architectural components. The results based on the MambaCNN baseline confirm the advantage of SNN-based reconstruction over a standard CNN decoder. On the PU dataset, MambaSNN obtains 91.40% mOA, outperforming MambaCNN (91.05%) and "Without SNN" (90.82%), while removing Mamba causes a more significant decline to 88.71%. These performance gradients indicate that while the SpeMamba module constitutes the primary source of architectural advantage, the SNNRecNet reconstruction branch contributes meaningful supplementary enhancements. The direct comparison between MambaSNN and MambaCNN specifically isolates the contribution of spiking-based reconstruction, revealing its superior effectiveness compared to conventional CNN approaches when the feature extraction backbone remains constant.

The similar performance patterns are observed on the IP dataset, where MambaSNN reaches 75.15% mOA, exceeding MambaCNN (74.94%), "Without SNN" (74.75%), and "Without Mamba" (74.37%). Despite the relatively limited scope of the absolute margins on the IP dataset, the consistent ranking validates the generalizability of the architectural design across diverse hyperspectral imaging scenarios. On the H dataset, MambaSNN achieves 89.90% mOA, substantially outperforming MambaCNN (88.16%, +1.74%) and "Without SNN" (87.84%, +2.06%), while "Without Mamba"

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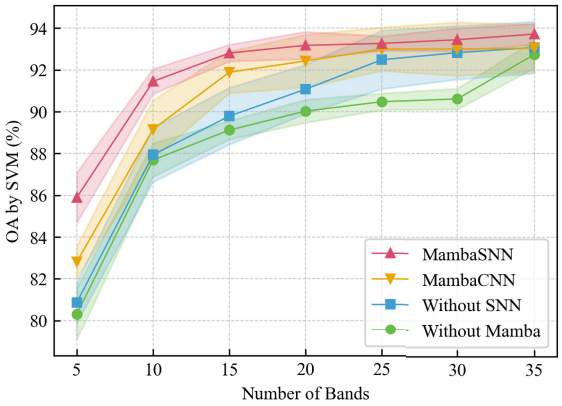


Fig. 15. OA results using the SVM classifier on PU dataset under a different number of selected bands in ablation analysis. The figure shows the mean and standard deviation of the results over ten runs.

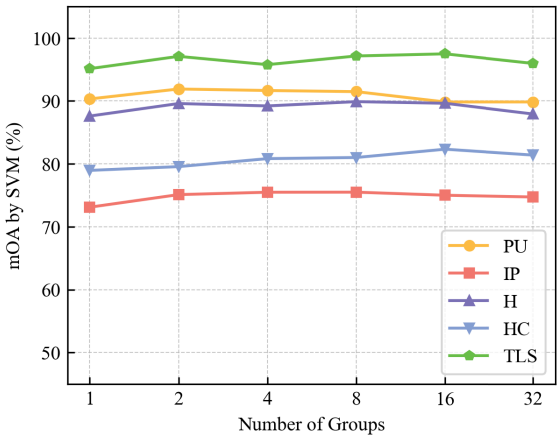


Fig. 16. Hyperparameter analysis of using the SVM classifier on five datasets.

exhibits the most significant decline (87.25%, +2.65%). This suggests that spiking reconstruction mechanisms provide enhanced benefits in complex urban environments where subtle spectral–spatial relationships are critical for effective BS. Consistent trends emerge on the HC and TLS datasets, where MambaSNN attains 82.33% and 97.49% mOA, respectively, systematically exceeding all ablated scenarios. The performance differentials between MambaSNN and MambaCNN (0.42% on HC, 0.41% on TLS) demonstrate that SNN-based reconstruction provides consistent advantages across diverse spatial resolutions and land cover compositions. These results collectively establish that the benefits of spiking reconstruction are not dataset-specific but represent fundamental architectural advantages.

Fig. 15 provides detailed insights into component behavior under varying spectral constraints through analysis of the PU dataset. Under severely limited spectral conditions (5 bands), removing the Mamba component causes dramatic performance degradation (5.58% OA reduction). It indicates that state-space modeling capabilities are essential for capturing long-range spectral dependencies when information is scarce. In contrast, modifying the reconstruction branch mainly affects performance stability and robustness rather than peak accuracy. Both "Without SNN" and "MambaCNN" exhibit more noticeable performance fluctuations across different band subsets and lower peak accuracy than the full model. Although MambaCNN provides a slight improvement over complete reconstruction removal, it cannot fully replicate the stability benefits inherent in spiking-based reconstruction. This finding suggests that the sparse activation patterns characteristic of SNNs contribute to more robust spectral representation learning.

The efficiency advantages of the complete architecture are particularly noteworthy. MambaSNN achieves 92.5% OA using only 15 spectral bands on the PU dataset, while ablated variants typically require 25–35 bands to attain comparable performance levels, representing a 67%–133% increase in spectral requirements. This efficiency enhancement has direct implications for practical deployment scenarios where storage, transmission, and computational resources are constrained. In addition, the full model demonstrates reduced variance across the entire range of selected bands, further confirming its robustness. These findings reinforce the importance of the Mamba module in extracting discriminative features from sparse spectral inputs, while the SNN module is vital for maintaining representational stability and spectral integrity. These components enhance the efficiency and reliability of BS, especially in scenarios where spectral resources are constrained.

F. Hyperparameter Analysis

A critical hyperparameter in the MambaSNN architecture is the spectral group cardinality G, which determines how the spectral dimension is partitioned. Fig. 16 presents the mOA values under SVM classification for various group cardinalities across all datasets. For the IP dataset, which has 200 bands over a broad 0.4–2.5 μm range, the performance reaches saturation at moderate group sizes. When G is set to 4 or 8, the resulting mOA values (75.49% and 75.50%, respectively) are very close. Further increases in G do not yield additional performance and instead lead to higher computational costs. Therefore, a G of 4 is selected as the preferred setting to balance accuracy and efficiency. In the case of the H dataset, which contains diverse urban and natural land classes with more intricate spectral patterns, setting G to 8 achieves near-maximum mOA (89.89%) while maintaining reasonable computational efficiency. For the HC and TLS datasets, characterized by higher spectral dimensionality (274 and 420 bands, respectively) and complex structures, the highest mOA (82.33% and 97.49%, respectively) is attained when G is set to 16. The smaller or larger values of G cause slight performance degradation. The PU dataset, with fewer bands and a narrower spectral range, shows stable performance across all tested G values. However, a noticeable decline in mOA is observed when G exceeds 8.

To further confirm the optimal G value for the PU dataset, we conducted an extensive hyperparameter study using 15 selected bands. Fig. 17 presents the violin plots for OA, AA, Kappa, and BS time based on this dataset. The results indicate that a G of 2 yields the highest median accuracy

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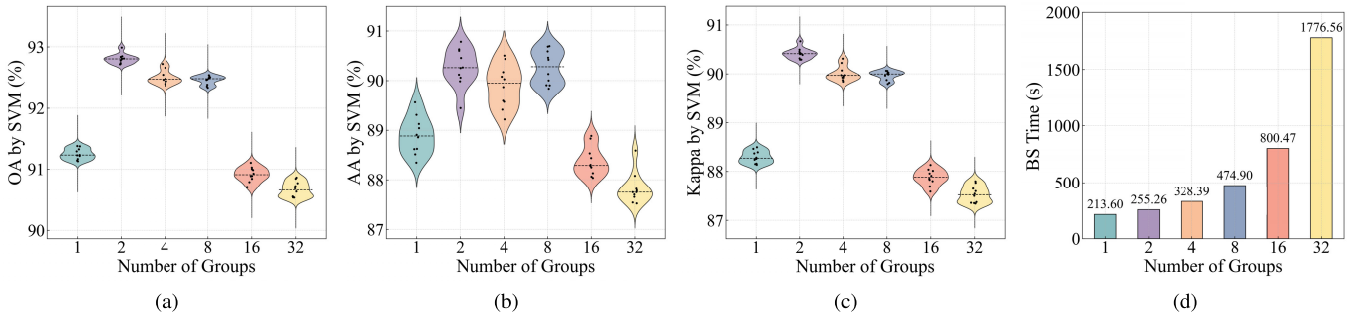


Fig. 17. Hyperparameter analysis of using the SVM classifier on the PU dataset. (a) OA. (b) AA. (c) Kappa. (d) BS time.

(OA = 92.78%, AA = 90.16%, and Kappa = 90.37%) with very narrow interquartile ranges, indicating strong stability and consistent behavior. The compact statistical distribution suggests that a smaller group size helps preserve local spectral continuity while enabling effective feature discrimination. When G exceeds 8, the classification performance degrades substantially. Specifically, the median OA decreases from 92.48% at a G of 8 to 85.20% when G is increased to 32, and the runtime increases from 255.26 to 1776.56 s. Excessive fragmentation limits the spectral information available to each group and weakens the representational capacity. These observations indicate that setting the G to 2 provides an effective tradeoff among spectral cohesion, discriminative ability, and computational efficiency on the PU dataset.

V. Conclusion

HSI can provide abundant spectral–spatial information, making it indispensable for tasks such as land cover classification and mineral exploration. In this context, BS emerges as a critical technique for reducing computational costs and alleviating information redundancy. This study presents MambaSNN, an unsupervised BS method for HSI-based land classification.

The main contributions and findings of this work are summarized as follows.

1) MambaSNN introduces a novel architecture that combines the SpeMamba and SNNRecNet modules to achieve both efficient spectral representation and robust band reconstruction. This framework leads to improvements in both classification accuracy and processing efficiency.
2) The SpeMamba module explicitly captures long-range spectral dependencies via recursive state transitions with linear complexity $\mathcal{O}(N)$. It enables scalable and informative feature extraction across the full spectral range. Meanwhile, the SNNRecNet module incorporates LIF neurons within a convolutional reconstruction network, promoting sparse feature encoding and noise resilience.
3) Extensive experiments conducted on the PU, IP, H, HC, and TLS datasets demonstrate that MambaSNN achieves superior performance in both BS accuracy and computational efficiency when compared with existing state-of-the-art methods. The proposed method demonstrates strong generalization ability and practical relevance, offering a computationally efficient solution for HSI analysis, especially suited for high-dimensional data processing in real-world scenarios.

Looking ahead, future research will focus on developing multiscale spectral feature extraction strategies to better capture both fine-grained vegetation cues and scenarios with highly homogeneous class semantics, especially under severely constrained BS conditions. Another promising direction is to explore spatial–spectral joint modeling and more adaptive BS mechanisms, so that the model can automatically determine compact yet informative band subsets for different datasets and application scenarios. Beyond these methodological extensions, the proposed framework also suggests broader research opportunities in efficient hyperspectral representation learning, lightweight neuromorphic remote sensing models, and robust unsupervised feature selection for resource-constrained applications.

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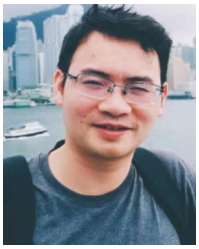
Zhenyu Wang received the B.S. degree in geographic information science from China University of Petroleum (East China), Qingdao, China, in 2023. He is currently pursuing the M.S. degree with the School of Geographic Sciences, Hunan Normal University, Changsha, China.

His research interests include deep learning and dimensionality reduction of spatial big data.



Zhongan Tang received the Ph.D. degree in geodesy and survey engineering from Wuhan University, Wuhan, China, in 2004.

He worked as a Researcher at the School of Construction and Environment, The Hong Kong Polytechnic University, Hong Kong, and carried out post-doctoral work at Central South University, Changsha, China. He is currently a Professorial Senior Engineer with The Third Surveying and Mapping Institute of Hunan Province, Changsha. His research interests include natural resources, big data mining, and uncertainty analysis.



Yiliang Wan received the B.S. degree in remote sensing science and technology and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2009 and 2015, respectively.

He is currently a Professor with the School of Geographic Sciences, Hunan Normal University, Changsha, China. His research interests include remote sensing image processing and applications, and spatial uncertainty modeling.



Wenzhong Shi received the Ph.D. degree from the University of Osnabruck in Vechta, Vechta, Germany, in 1994.

He is currently the Director of Otto Poon Charitable Foundation Smart Cities Research Institute, PolyU, Hong Kong, the Director of PolyU-Shenzhen Technology and Innovation Research Institute (Futian), Shenzhen, China, the Chair Professor in geographic science and remote sensing, and the Director of the Joint Research Laboratory on Spatial Information of PolyU and Wuhan University, Wuhan, China. He is an Academician of the International Eurasian Academy of Sciences. His research interests include GISci, remote sensing, and urban informatics, with a focus on analytics and quality control for spatial big data, object extraction, and change detection from satellite images and LiDAR data, integrated indoor mapping technology, 3-D and dynamic GISci modeling, and smart city applications.

Dr. Shi is a fellow of the Academy of Social Sciences, U.K., and the Royal Institution of Chartered Surveyors and Hong Kong Institute of Surveyors. He serves as the President of the International Society for Urban Informatics and the Editor-in-Chief of the international journal *Urban Informatics*.



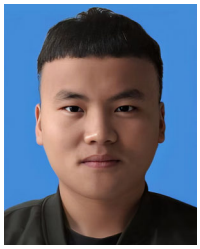
Hua Zhang received the Ph.D. degree in cartography and geographical information engineering from China University of Mining and Technology, Xuzhou, China, in 2012.

He is currently a Professor of GIS and remote sensing with the School of Environment and Spatial Informatics, China University of Mining and Technology. His research interests include multispectral/hyperspectral and high-resolution remotely sensed image processing, uncertainty in classification, and remote sensing applications.



Chunhui Li received the M.S. degree in geomatics engineering from Central South University, Changsha, China, in 2014.

She currently serves as a Researcher with The Third Surveying and Mapping Institute of Hunan Province, Changsha. Her research interests include geographic information security, geographic information processing, and big data mining.



Yang Zhao received the M.S. degree in cartography and geographic information systems from Hunan Normal University, Changsha, China, in 2023, where he is currently pursuing the Ph.D. degree in cartography and geographic information systems with the School of Geographic Sciences.

His research interests include spatiotemporal data mining and urban remote sensing.

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